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Machine Learning for collapse structural analysis of Hollow Elements

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For those with the talent, the will, but not the opportunity.

Abstract

The prediction of the collapse of hollow structural elements is traditionally based on nonlinear finite element (FE) analyses, which are accurate but computationally expensive, making them unsuitable for extensive parametric studies or rapid assessments. This thesis develops an automated framework that replaces a large part of the numerical simulations with surrogate models based on machine learning.

The workflow integrates Sobol sampling for the Design of Experiments, high-fidelity Finite Element Model (FEM) simulations in Abaqus, automatic extraction of collapse indicators (Area Reduction, Diameter Reduction) and vectorial responses (Area Profile), and finally the construction of surrogate models using Gaussian Process Regression (GPR) and Multi-Layer Perceptrons (MLP).

The results show that the surrogates reproduce the reference simulations with engineering-level accuracy, reducing computation times from 80–700 seconds per run to approximately 0.1 seconds per prediction, achieving a speed-up of three to four orders of magnitude. GPR also provides uncertainty quantification, useful for engineering applications, while MLP proves more suitable for larger and more complex datasets.

Although the work is based on simulated data and has room for improvement related to numerical stabilization and uncertainty calibration, it provides a validated, extensible, and reproducible tool. This opens the way for future developments such as sequential sampling, hybrid model approaches, with potential impact on design and predictive maintenance.

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Listing of acronyms

AP ..... Area Profile

AR ..... Area Reduction

CV ..... Cross-Validation

DoE ..... Design of Experiments

DR ..... Diameter Reduction

FEA ..... Finite Element Analysis

FEM ..... Finite Element Method

GBDTR ..... Gradient Boosting Decision Tree

GP ..... Gaussian Process

GPR ..... Gaussian Process Regression

KNNR ..... K-Nearest Neighbors

LHS ..... Latin Hypercube Sampling

LOO ..... Leave-One-Out

LOOCV ..... Leave-One-Out Cross-Validation

MAE ..... Mean Absolute Error

ML ..... Machine Learning

MLP ..... Multi-Layer Perceptron

MSE ..... Mean Squared Error

NN ..... Neural Networks

PCHIP ..... Piecewise Cubic Hermite Interpolating Polynomial

PDE ..... Partial Differential Equation

PINN ..... Physics-Informed Neural Network

QoI ..... Quantity of Interest

RAM ..... Random Access Memory

RBF ..... Radial Basis Function

RMSE ..... Root Mean Squared Error

S4R ..... Shell 4-node Reduced integration element

SVM ..... Support Vector Machine

SVR ..... Support Vector Regression

1 Introduction

The analysis of structural collapse in thin-walled hollow elements represents a long-standing challenge in computational mechanics. While nonlinear Finite Element Analysis (FEA) provides the reference methodology for capturing the combined effects of material plasticity and geometric instability, its prohibitive computational cost limits its use in large-scale parametric studies, real-time assessment, and design optimization. The increasing demand for predictive tools that are both accurate and computationally efficient has therefore motivated the integration of data-driven approaches into traditional simulation workflows.

This thesis addresses the problem by constructing a surrogate modeling framework that substitutes the majority of high-fidelity finite element simulations with machine learning metamodels. The pipeline is designed to be fully automated, beginning with the generation of a representative dataset through a Design of Experiments strategy, continuing with the extraction of collapse indicators and deformed profiles, and culminating in the training and validation of regression models. Gaussian Process Regression (GPR) and Multi-Layer Perceptron (MLP) are investigated as primary candidates, selected for their complementary strengths in uncertainty quantification and nonlinear function approximation.

The work aims to demonstrate that such surrogate models are capable of predict-

ing both scalar quantities of interest and vector-valued structural responses with accuracy sufficient for engineering applications. By achieving computational speedups of several orders of magnitude while retaining predictive reliability, the proposed framework establishes a foundation for robust design exploration, uncertainty-aware decision-making, and transferability to other domains characterized by expensive high-fidelity simulations.

2

Theoretical and Methodological Background

This chapter builds the theoretical foundations necessary for the development of the surrogate model. It begins with the physical Solid Mechanics formulation of the problem, introduces its standard high-fidelity numerical approach, and highlights the computational limitations. This motivate the need for an efficient sampling of the parameter space and a meta model. In particular, this chapter treat Gaussian Processes models, comparing them with other alternatives.

2.1 Solid Mechanics and Structural Models

2.1.1 Kinematics and strain measures

A continuous body is described by a displacement field $\boldsymbol{u}(\boldsymbol{X}, t)$ defined in the reference configuration Ω_0. The variables $\boldsymbol{x}, \boldsymbol{X}$ represent the positions of the continuous body points in the deformed (spatial) and reference (material) configuration. The transformation $\boldsymbol{X}$ to $\boldsymbol{x}$ is given by

$$\boldsymbol{x}(\boldsymbol{X}) = \boldsymbol{X} + \boldsymbol{u}(\boldsymbol{X}).$$

The deformation gradient is

$$\boldsymbol{F} = \frac{\partial \boldsymbol{x}}{\partial \boldsymbol{X}}, \qquad J = \det \boldsymbol{F},$$

which maps vectors from the reference configuration to the current one. To quantify deformation, several strain measures can be used:

$$\boldsymbol{\epsilon} = \tfrac{1}{2}\left(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^\top\right) \qquad \text{(linear strain)}, \tag{2.1}$$

$$\boldsymbol{E} = \tfrac{1}{2}\left(\boldsymbol{F}^\top \boldsymbol{F} - \boldsymbol{I}\right) \qquad \text{(Green–Lagrange strain tensor)} \quad [1]. \tag{2.2}$$

The first is a linearization valid for small deformations, while the second is suitable for large displacements and rotations. Given the expected large displacements and rotations associated with the Hollow Element's instability phenomena (buckling), the linearized tensor $\boldsymbol{\epsilon}$ is inadequate. Therefore, the Green–Lagrange strain tensor $\boldsymbol{E}$ is considered.

2.1.2 Equilibrium equations

The quasi-static balance of forces in the reference configuration Ω_0 can be written in strong form as:

$$\mathrm{Div}\boldsymbol{P} + \boldsymbol{B} = \boldsymbol{0} \quad \text{in } \Omega_0,$$

with $\boldsymbol{P} = \boldsymbol{F}\boldsymbol{S}$ being the first Piola–Kirchhoff stress tensor, $\boldsymbol{S}$ the second Piola–Kirchhoff stress tensor, and $\boldsymbol{B}$ the body forces. This material (or Lagrangian) formulation is adopted because the domain of integration Ω_0 remains fixed throughout the deformation. The boundary conditions are:

$$\boldsymbol{u} = \bar{\boldsymbol{u}} \text{ on } \Gamma_u, \qquad \boldsymbol{P}\boldsymbol{N} = \bar{\boldsymbol{t}} \text{ on } \Gamma_t,$$

where Γ_u and Γ_t are subsets of the boundary. Since the strong form imposes regularity requirements on the displacement field that are difficult to satisfy in complex problems and does not directly permit numerical discretization, its integral counterpart (weak form) is introduced. By multiplying by a virtual field $\boldsymbol{w}$ and integrating

by parts, the weak form is obtained:

$$\int_{\Omega_0} \boldsymbol{S} : \delta \boldsymbol{E}\, dV = \int_{\Omega_0} \boldsymbol{B} \cdot \boldsymbol{w}\, dV + \int_{\Gamma_t} \bar{\boldsymbol{t}} \cdot \boldsymbol{w}\, dA,$$

which forms the basis for the Finite Element Method [2, 3].

2.1.3 Constitutive laws

For a consistent finite-strain formulation, the constitutive model is based on the multiplicative decomposition of the deformation gradient into elastic and plastic parts [4]:

$$\boldsymbol{F} = \boldsymbol{F}^e \boldsymbol{F}^p.$$

Here, $\boldsymbol{F}^p$ represents the irreversible plastic flow, mapping to a stress-free intermediate configuration, while $\boldsymbol{F}^e$ accounts for the subsequent elastic deformation and rotation to the current configuration. Plastic flow in metals is assumed to be isochoric, which implies the constraint $J^p = \det(\boldsymbol{F}^p) = 1$.

Elastic response. The elastic behavior is described as hyperelastic, with stress derived from a stored-energy function $W(\boldsymbol{C}^e)$ that depends on the elastic right Cauchy–Green tensor $\boldsymbol{C}^e = (\boldsymbol{F}^e)^\top \boldsymbol{F}^e$. The second Piola–Kirchhoff stress tensor $\boldsymbol{S}$ in the reference configuration is obtained by pulling back the stress from the intermediate configuration:

$$\boldsymbol{S} = (\boldsymbol{F}^p)^{-1} \left(2 \frac{\partial W(\boldsymbol{C}^e)}{\partial \boldsymbol{C}^e} \right) (\boldsymbol{F}^p)^{-\top}.$$

Rate decomposition. The spatial velocity gradient $\boldsymbol{l} = \dot{\boldsymbol{F}} \boldsymbol{F}^{-1}$ is additively decomposed into elastic and plastic contributions, $\boldsymbol{l} = \boldsymbol{l}^e + \boldsymbol{l}^p$. Its symmetric part, the rate of deformation $\boldsymbol{d} = \mathrm{sym}(\boldsymbol{l})$, is decomposed likewise as $\boldsymbol{d} = \boldsymbol{d}^e + \boldsymbol{d}^p$.

Yield criterion. The von Mises yield function is expressed in terms of the Kirchhoff stress $\boldsymbol{\tau} = J\boldsymbol{\sigma}$ to ensure objectivity:

$$f(\boldsymbol{\tau}, \kappa) = \|\mathrm{dev}(\boldsymbol{\tau})\| - \sqrt{\tfrac{2}{3}}\, \sigma_y(\kappa) \leq 0,$$

where $\mathrm{dev}(\boldsymbol{\tau})$ is the deviatoric part of $\boldsymbol{\tau}$. The yield stress σ_y is defined by a piecewise-linear isotropic hardening function of the equivalent plastic strain κ, which takes the linear form:

$$\sigma_y(\kappa) = \sigma_{y0} + H\kappa,$$

where σ_{y0} is the initial yield stress and H is the constant hardening modulus. This function is specified by the initial yield stress and a second point on the stress-strain curve, consistent with the bilinear material model implemented in the FEM analysis (see Section 3.4.1). The hardening modulus H is therefore calculated from the model input parameters as:

$$H = \frac{\mathrm{SY2} - \mathrm{SY1}}{\mathrm{EP2}},$$

where SY1 is the initial yield stress (σ_{y0}), and (SY2, EP2) are the coordinates of the second point defining the hardening curve.

Flow rule and hardening. Plastic evolution is governed by the Kuhn–Tucker conditions,

$$f \leq 0, \quad \dot{\gamma} \geq 0, \quad \dot{\gamma} f = 0,$$

and the associated flow rule for the plastic rate of deformation,

$$\boldsymbol{d}^p = \dot{\gamma} \, \frac{\partial f}{\partial \boldsymbol{\tau}}.$$

Integration of this constitutive model is performed through incremental return-mapping schemes with a consistent algorithmic tangent operator, as detailed in [4].

This elasto-plastic formulation enables the correct representation of dissipated energy and irreversible stress redistribution during post-buckling collapse, ensuring that the data for the surrogate model carries information about the essential features of instability phenomena.

2.2 The Finite Element Method (FEM)

2.2.1 Discretization

The FEM approximates the displacement field as a combination of shape functions $N_a(\xi, \eta)$ over the elements:

$$\mathbf{u}^h(\mathbf{x}) = \sum_{a=1}^{n_{en}} N_a(\mathbf{x})\, \mathbf{u}_a,$$

where $\mathbf{u}_a$ are the nodal displacements. Substituting this approximation into the weak form yields the discrete system

$$\mathbf{R}(\mathbf{q}) = \mathbf{F}^{\text{int}}(\mathbf{q}) - \mathbf{F}^{\text{ext}} = \mathbf{0},$$

which in the linear regime reduces to the classic matrix form

$$\mathbf{K}\mathbf{q} = \mathbf{f}.$$

In nonlinear problems, $\mathbf{R}(\mathbf{q})$ is solved iteratively using the Newton–Raphson method:

$$\mathbf{K}_t(\mathbf{q}_i)\Delta\mathbf{q}_i = -\mathbf{R}(\mathbf{q}_i), \qquad \mathbf{q}_{i+1} = \mathbf{q}_i + \Delta\mathbf{q}_i,$$

where $\mathbf{K}_t$ is the consistent tangent matrix.

2.2.2 Shell elements for thin structures

For structures with $thk/D \ll 1$ (thin walls), the use of three-dimensional solid elements is inefficient. Therefore, reduced models are adopted:

- **Kirchhoff–Love**: valid for very thin shells, they neglect transverse shear.
- **Mindlin–Reissner**: they include transverse shear, making them more suitable for moderately thick shells.

For this study, elements based on Mindlin-Reissner kinematics are adopted, which can model transverse shear deformation, a non-negligible effect in the post-buckling regime. Specifically, 4-node quadrilateral shell elements with reduced integration

(S4R shell element in Abaqus notation) are used. This formulation offers an optimal trade-off between accuracy and computational cost, while also mitigating numerical locking phenomena [1, 5].

2.2.3 Nonlinearity and instability

There are two main sources of nonlinearity:

1. **Geometric**: large displacements and rotations, instability effects (buckling).
2. **Material**: plasticity and hardening.

Buckling can be studied in two steps:

- *Linear eigenvalue analysis*, by solving

$$\Big(\mathbf{K}_{\text{mat}} + \lambda \mathbf{K}_{\text{geo}}\Big)\phi = 0,$$

  which provides critical loads λ and mode shapes ϕ. [6]

- *Nonlinear post-buckling analysis*, which requires incremental and path-following techniques, such as the arc-length method (Riks) or stabilized *Static General* solver [7, 8].

Linear eigenvalue analysis is employed to identify the critical loads and the most likely instability mode shapes. These mode shapes are subsequently used to define the initial geometric imperfections in the nonlinear model. This is a standard procedure to trigger a realistic post-buckling equilibrium path, which is then traced using nonlinear analysis.

2.2.4 Relevant numerical aspects

In shell models, certain numerical issues must be addressed. The adoption of shell elements with Mindlin-Reissner kinematics introduces the risk of Shear Locking for very thin shells. This problem is mitigated by using reduced integration, a feature of the S4R shell element [1, 2]. Reduced integration, in turn, can generate spurious zero-energy modes, known as Hourglass Modes [1]. The software implementation of these elements includes numerical stabilization schemes to control these non-physical instabilities. Furthermore, mesh sensitivity studies were conducted to find

an effective compromise for the mesh size that would not generate spurious stresses while maintaining an acceptable computational speed.

The combination of a finite-strain kinematics, an elasto-plastic constitutive model, a shell element discretization, and an advanced nonlinear solver allows for the construction of a high-fidelity numerical model capable of simulating the collapse of a thin-walled Hollow Element. Although accurate, the computation time for a single simulation is often on the order of hours, making an exhaustive exploration of the parameter space, which would require thousands of such analyses, computationally prohibitive. This intractability of the FE model necessitates the use of a strategy for the intelligent selection of a limited number of numerical experiments. This strategy is the subject of the Design of Experiments (DoE) discipline, discussed in the following section.

2.3 Design of Experiments (DoE) for Simulations

Design of Experiments (DoE) is a statistical discipline for planning, conducting, and analyzing experiments, both physical and computational, to extract the maximum amount of information with the minimum expenditure of resources. In the context of engineering and numerical simulation (e.g., Finite Element Analysis, FEA), where a single model evaluation can take hours or days of computation, DoE is a fundamental tool for the efficient exploration of the input parameter space.

The primary goal is not to perform an exhaustive, computationally prohibitive analysis, in all the parameter domain, but to select a small and strategic subset of input combinations. This subset must be sufficiently representative to capture the global behavior of the system and the interactions between variables.

The Curse of Dimensionality: A fundamental challenge in exploring parameter spaces is the so-called "curse of dimensionality." As the number of input variables (the dimensions) increases, the volume of the space grows exponentially. Consequently, the number of points required to sample this space with a given density also grows exponentially, rendering any attempt at exhaustive sampling computationally prohibitive. This phenomenon underscores the impossibility of densely

"filling" a high dimensional space and necessitates intelligent sampling strategies that can extract meaningful information from a limited number of simulations.

The quality and accuracy of a surrogate model are intrinsically linked to the quality and distribution of the training data. This is where DoE plays a crucial role: the table of experiments generated by the DoE constitutes the training dataset for the metamodel. A well designed DoE ensures that the sampling points are distributed to explore the entire domain of interest, avoiding clustering in some areas and empty zones in others, thereby maximizing the predictive capability of the resulting surrogate.

Before analyzing individual techniques, it is essential to establish the fundamental principles that distinguish the design of computer experiments from its classical counterpart, developed for physical experiments. This paradigm shift is key to understanding why entirely new methodologies have been developed.

2.3.1 Physical Experiments and Computer Experiments

The most profound difference between "classical" and "modern" DoE lies in the nature of experimental error. Physical experiments are inherently stochastic. Even when attempting to control all known variables, there are always sources of uncontrolled or unknown variability (e.g., temperature fluctuations, material impurities, measurement errors) that introduce random error into the output. Consequently, classical DoE is founded on three pillars to manage this randomness: replication, to obtain a reliable estimate of the experimental error; randomization of the experiment order, to mitigate the effect of uncontrolled confounding variables; and blocking, to group experiments into homogeneous conditions and reduce variability. The primary objective of these designs is often the efficient estimation of the coefficients of a predefined statistical model, typically linear or quadratic. To minimize the variance of these estimates, the optimal sampling points are often placed at the vertices and on the faces of the factor space, where the statistical leverage is greatest.[9] In contrast, computer experiments using deterministic models doesn't have any random error. Running the same simulation with the exact same inputs will always produce the exact same output.[10] This characteristic makes replication useless and randomization of the execution order irrelevant. The objective is no longer to estimate the parameters of an a priori model, but rather to approximate a

complex and potentially highly nonlinear ”black-box” function, which is the simulation itself. This radically shifts the emphasis from parameter estimation to function approximation over the entire domain of interest. The question is not ”what is the effect of factor A?” but ”how can I build an accurate map of the response surface generated by all factors?”. This change in objective is the driving force behind the need to sample the interior of the domain and not just its boundaries, leading directly to the concept of ”space-filling.”

Given the deterministic nature of simulations, two sampling points that are very close in the input space will provide probably very similar outputs, resulting in largely redundant information. The guiding philosophy of modern DoE thus becomes to distribute the sampling points in such a way as to ”fill” the design space as uniformly and completely as possible, maximizing the distance between points and minimizing the presence of large unexplored regions. This approach is known as space-filling.[11]

A good space-filling design ensures that, wherever one is in the domain, there is a relatively close data point, thus providing a solid basis for interpolation and the construction of an accurate global surrogate. The quality of the space-filling is measured through various criteria, which can be grouped into two main categories [9]:

- Distance-based criteria: These criteria evaluate the distribution of points based on the Euclidean distances between them.
- Uniformity-based (or discrepancy) criteria: These criteria measure how much the distribution of points deviates from a perfectly uniform distribution.

DoE techniques for computer experiments can be classified into two broad strategic families: static and adaptive. Static (or ”One-Shot”) Strategies: In this approach, the entire set of experimental points is defined in a single step, before any simulations are run.[9] These strategies are inherently exploratory; in the absence of information about the response function, the best bet is to cover the entire domain as uniformly as possible. They are ”agnostic” to the response and form the basis for building the initial surrogate model. Classic examples include Latin Hypercube Sampling (LHS) and low-discrepancy sequences.

Adaptive (or Sequential) Strategies: In this approach, sampling points are selected and added iteratively.[11] One starts with a small initial static design, builds

a preliminary surrogate model, and then uses this model to intelligently decide where to place the next point. This introduces a feedback loop that allows for a balance between two competing objectives:

- Exploration: Sampling in regions where the surrogate model's uncertainty is high, in order to improve its global accuracy and have an extended domain.
- Exploitation: Sampling in regions where the surrogate model predicts promising values, in order to converge quickly to an optimal solution.

This duality makes adaptive strategies extremely powerful and efficient in terms of computational budget, especially when the final goal is optimization rather than simply mapping the response.

2.3.2 Static sampling techniques

Static sampling techniques are the starting point for most surrogate modeling applications. Their goal is to provide an initial database that efficiently covers the entire parameter space.

Latin Hypercube Sampling (LHS)

Latin Hypercube Sampling (LHS) is one of the most popular and widely used DoE techniques for computer experiments, first introduced in a formal context by McKay, Beckman, and Conover in 1979.[12]

LHS is a form of stratified sampling that ensures full coverage of the range of each individual input variable. To generate an LHS design with N samples in a k-dimensional space, the procedure is as follows:

- For each of the k variables, its range of variation is divided into N intervals of equal probability.
- From each interval of each variable, a single value is drawn.
- The sampled values for the k variables are then randomly combined to form the N sampling points (input vectors).

The Latin Hypercube rule dictates that if the N-dimensional design is projected onto any axis for a single variable, there is exactly one point in each of the N

intervals.[9] The main strength of LHS is its excellent one-dimensional projection property. It guarantees that each variable is sampled across its entire range, avoiding the clustering that can occur with simple random sampling (Monte Carlo). Furthermore, its construction is computationally simple and independent of the number of dimensions.[12] [13]

Although LHS ensures good marginal coverage, the random combination of values can still lead to a poor distribution of points in the multidimensional space. It is possible for clusters of points to form, leaving large regions of the design space uncovered.[10] For this reason, many possible LHS designs are often generated, and the best one is selected according to a secondary space-filling criterion, such as maximizing the minimum distance between points (Maximin LHD) or minimizing the potential energy of the system of points. This optimization process creates hybrid designs that combine the good projective properties of LHS with better global space-filling properties. These approaches are known as hybrid designs, as they combine the stratified structure of one technique (LHS) with a geometric optimization criterion (Maximin) to inherit the strengths of both methodologies.

Sobol quasi-random sequence

The Sobol sequence is a low-discrepancy sequence of points designed to uniformly sample an s-dimensional unit hypercube $I^s = [0, 1]^s$. It is used in Quasi-Monte Carlo methods for integration and optimization, ensuring a convergence rate of order $O(N^{-1}(\log N)^s)$, which is superior to that of pseudo-random methods. [14]

The construction is based on binary operations ($\oplus$ represents the bit-wise XOR operation). The construction is based on a series of logical steps that use binary arithmetic.

- Direction Numbers (v_i): For each dimension, a primitive polynomial over $\mathbb{Z}_2$ is chosen, defining a recurrence relation to generate a sequence of binary fractions v_i, called direction numbers.

- Sequence Points (x^n): The n-th point of the sequence is the result of the XOR operation between the direction numbers v_i for which the i-th bit in the binary representation of n is 1.

A more efficient recursive calculation (Antonov-Saleev) uses the Gray code of the

index n. The next point is obtained from the previous one by:

$$x^{n+1} = x^n \oplus v_c$$

where c is the index of the least significant zero-bit of n.

For sampling in s dimensions, s distinct primitive polynomials are used, each generating an independent sequence for each component of the vector $q^n = (x_1^n, x_2^n, \ldots, x_s^n)$.

Figure 2.1 visually illustrates the qualitative difference between the main static sampling strategies.

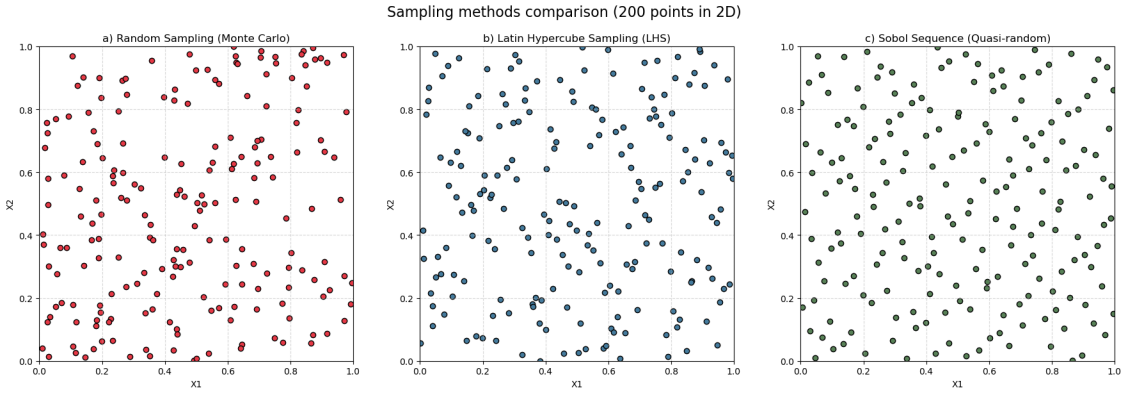


Figure 2.1: Visual comparison with N=200 points in a 2D domain between: (a) Simple Random Sampling, showing clusters and empty zones; (b) Latin Hypercube Sampling (LHS), with uniform projections on the axes but possible aggregations; (c) Sobol Sequence, showing the best space-filling properties.

Distance-Based Designs: The Maximin and Minimax Criteria

An alternative approach to generating *space-filling* designs is to formulate the problem as a numerical optimization based on distance metrics. The two most well-known criteria were proposed by Johnson, Moore, and Ylvisaker in 1990.

Principle: The idea is to place the points to optimize a certain geometric property of their configuration [10].

Maximin Criterion: This criterion seeks the design that maximizes the minimum distance between any pair of points. Intuitively, it "pushes" the points as far away from each other as possible to avoid clustering. Its mathematical formulation is:

$$\max_{D} \min_{i \neq j} d(\mathbf{x}_i, \mathbf{x}_j)$$

Minimax Criterion: This criterion seeks the design that minimizes the maximum distance from any point in the space to the nearest sampled point. The goal is to ensure there are no large empty "holes" in the domain. Its formulation is:

$$\min_{D} \max_{\mathbf{x} \in \mathcal{X}} \min_{i} d(\mathbf{x}, \mathbf{x}_i)$$

Constructing these designs is a non-convex global optimization problem and is computationally very demanding, especially in high dimensions. Moreover, purely maximin designs, despite having excellent separation, can suffer from poor projection properties, with points tending to collapse onto a few levels. One of the most effective solution is often a hybrid approach, such as finding a *Maximin LHD*, a Maximin design within the class of Latin Hypercube Designs.

Strategy adopted The choice of the optimal static sampling strategy is not universal but depends strictly on the nature of the problem and the type of surrogate model to be built. A sampling with good separation properties like Sobol Sampling, for example, is particularly suitable for interpolation-based surrogates such as Kriging, which benefit from well-distributed data points.

2.3.3 Dynamic sampling techniques

While static strategies provide an excellent foundation for global exploration, adaptive strategies offer a way to intelligently refine the surrogate model, focusing computational resources where they are most needed. This approach transforms Design of Experiments (DoE) from a purely statistical problem into a sequential decision theory problem: at each step, one must decide which is the next most informative experiment to run, given the knowledge accumulated so far. It is very useful in *optimization problems*, where a minimum point is found.

The Principle of Expected Improvement (EI) and the EGO Algorithm

The most influential and widespread approach in adaptive sampling is based on the *Expected Improvement (EI)* criterion, formalized in the context of Efficient Global Optimization (EGO) by Jones, Schonlau, and Welch in 1998.

Probabilistic Surrogates (Kriging/Gaussian Processes): The fundamental prerequisite for effective adaptive sampling is the use of a surrogate model that provides not only a point prediction but also a measure of its associated uncertainty. The Kriging model, or its machine learning equivalent, the Gaussian Process (GP), is the tool of choice for this task [15]. A trained GP model provides for any new point $\mathbf{x}$ a probability distribution (Gaussian) for the function's value, characterized by a mean (the prediction) and a variance (the uncertainty). The uncertainty is typically low near already sampled points and high far from them.

The EI Acquisition Function: The EI criterion leverages both the prediction and the uncertainty of the GP model to guide the selection of the next point. Let $y_{\min}$ be the minimum value, the best choice for our optimization problem observed so far. For a new candidate point $\mathbf{x}$, the "improvement" over $y_{\min}$ is defined as:

$$I(\mathbf{x}) = \max(0, y_{\min} - Y(\mathbf{x}))$$

where $Y(\mathbf{x})$ is the random variable describing the function's value at $\mathbf{x}$ according to the GP model. Since $Y(\mathbf{x})$ is a distribution, its expected value, the Expected Improvement $\mathbb{E}[I(\mathbf{x})]$, is calculated. This quantity, which can be computed analytically, takes high values in two types of regions:

- Regions where the GP's mean prediction is low (lower than $y_{\min}$). This is the exploitation element, were a refined optimum point is reached.
- Regions where the GP's uncertainty (variance) is high. This is the exploration element, to find other different optima.

The EI function thus naturally and automatically balances the need to search near known good candidates and the need to explore unknown regions of the space.

The EGO (Efficient Global Optimization) Algorithm: The EGO algorithm is the procedural framework that implements this strategy [16]:

- Initialization: Select a small initial set of points using a static sampling technique (e.g., LHD) and run the expensive simulations.
- Modeling: Build a Gaussian Process (Kriging) surrogate model using all collected data.
- Acquisition Optimization: Find the point $\mathbf{x}^*$ that maximizes the EI acquisition function. This is a fast auxiliary optimization, based on the GP model and not the original simulation.
- Evaluation: Run the expensive simulation at the point $\mathbf{x}^*$ to obtain its true value.
- Iteration: Add the new (input, output) pair to the dataset and update the model, until the computational budget is exhausted.

Beyond Expected Improvement: Although EI is the most common acquisition function, it is not the only one. Other metrics such as the Probability of Improvement (PI), which maximizes the probability of improving upon the current minimum, or the Upper Confidence Bound (UCB), which favors points with high uncertainty, offer different balances between exploitation and exploration. Furthermore, if the goal is not optimization but the reduction of the surrogate's global error, adaptive strategies can be adopted whose purpose is to sample at points of maximum predictive variance of the model, regardless of the predicted mean value.

2.4 Gaussian Process (GP) Metamodeling

As highlighted in the previous sections, simulating the nonlinear collapse of thin structures requires high-fidelity FEM models. Although accurate, these are computationally expensive, to the point of making an extensive exploration of the design parameter space impracticable [17]. To overcome this limitation, metamodeling, or surrogate modeling, techniques are employed. The goal is to construct a mathematical approximation of the complex input-output relationship of the simulation, which is, however, much more efficient to query.

Among the various available techniques, this work relies on Gaussian Process Regression (GPR), a powerful non-parametric Bayesian inference method [18]. Also known in the field of geostatistics as *Kriging*, this approach is distinguished by the fact that it does not merely provide a point prediction but also accompanies it with a quantitative measure of uncertainty. This feature is of fundamental importance in engineering contexts, where evaluating the reliability of a prediction is crucial.

2.4.1 Theoretical Foundations of Gaussian Processes

A Gaussian Process (GP) can be understood as a generalization of the multivariate Gaussian distribution, extended to an infinite-dimensional function space. Formally, a GP is a collection of random variables, any finite subset of which has a joint Gaussian distribution [19].

A GP is completely defined by two elements: a mean function $m(\boldsymbol{x})$ and a covariance function, or kernel, $k(\boldsymbol{x}, \boldsymbol{x}')$:

$$f(\boldsymbol{x}) \sim \mathcal{GP}(m(\boldsymbol{x}), k(\boldsymbol{x}, \boldsymbol{x}')) \tag{2.3}$$

where:

- The mean function $m(\boldsymbol{x}) = \mathbb{E}[f(\boldsymbol{x})]$ represents our prior expectation of the function's behavior. For simplicity and without loss of generality, a zero mean, $m(\boldsymbol{x}) = 0$, is often assumed, leaving it to the kernel to capture the structure of the problem.
- The covariance function $k(\boldsymbol{x}, \boldsymbol{x}') = \mathbb{E}[(f(\boldsymbol{x}) - m(\boldsymbol{x}))(f(\boldsymbol{x}') - m(\boldsymbol{x}'))]$ is the core of the GP. It defines the correlation between the function's values at two distinct points, $\boldsymbol{x}$ and $\boldsymbol{x}'$, and encodes our prior assumptions about the function itself, such as its smoothness. Intuitively, the kernel implements the idea that points close in the input space should have similar outputs.

2.4.2 Gaussian Process Regression

The GPR framework leverages Bayes' theorem to update the prior distribution over functions in light of observed data, thus obtaining a posterior distribution that can be used to make predictions at new points.

Starting with a training dataset $\mathcal{D} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^n$, a model is assumed where the observation y_i is related to the latent function $f(\boldsymbol{x}_i)$ by Gaussian noise, $y_i = f(\boldsymbol{x}_i) + \varepsilon_i$, with $\varepsilon_i \sim \mathcal{N}(0, \sigma_n^2)$. In the context of computer experiments, which are deterministic, this noise term is theoretically absent. However, it is retained as a numerical regularization term, known as a *nugget*, to improve the stability of the calculation.

The posterior predictive distribution for a new set of test points $\boldsymbol{X}_*$, conditioned on the training data, is also a Gaussian, $\mathcal{N}(\bar{\boldsymbol{f}}_*, \text{cov}(\boldsymbol{f}_*))$. The mean and covariance of this predictive distribution can be calculated analytically [17, 19]:

$$\bar{\boldsymbol{f}}_* = \boldsymbol{K}(\boldsymbol{X}_*, \boldsymbol{X})[\boldsymbol{K}(\boldsymbol{X}, \boldsymbol{X}) + \sigma_n^2 \boldsymbol{I}]^{-1} \boldsymbol{y} \tag{2.4}$$

$$\text{cov}(\boldsymbol{f}_*) = \boldsymbol{K}(\boldsymbol{X}_*, \boldsymbol{X}_*) - \boldsymbol{K}(\boldsymbol{X}_*, \boldsymbol{X})[\boldsymbol{K}(\boldsymbol{X}, \boldsymbol{X}) + \sigma_n^2 \boldsymbol{I}]^{-1} \boldsymbol{K}(\boldsymbol{X}, \boldsymbol{X}_*) \tag{2.5}$$

Equation (2.4) provides the point prediction (the expected value), while the diagonal terms of the covariance matrix (2.5) represent the predictive variance at each point, from which confidence intervals are derived.

2.4.3 The Covariance Function (Kernel)

The choice of the kernel is a crucial step, as it defines the properties of the function that the GP can model. Different kernels correspond to different prior assumptions.

- **Squared Exponential (or RBF):** $k_{SE}(\boldsymbol{x}, \boldsymbol{x}') = \sigma_f^2 \exp\left(-\frac{\|\boldsymbol{x}-\boldsymbol{x}'\|^2}{2l^2}\right)$. This very common kernel produces infinitely differentiable, and therefore very smooth, functions. Its hyperparameters σ_f^2 (signal variance) and l (length scale) control the vertical amplitude of the oscillations and the "distance" over which points remain correlated, respectively.

- **Matérn family:** This family of kernels generalizes the RBF by introducing an additional parameter ν that controls the smoothness of the function. Common values such as $\nu = 3/2$ and $\nu = 5/2$ generate once and twice differentiable functions.

  $$k(x_i, x_j) = \sigma^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\sqrt{2\nu}\frac{\|x_i - x_j\|}{l}\right)^{\nu} K_\nu \left(\sqrt{2\nu}\frac{\|x_i - x_j\|}{l}\right)$$

  This flexibility often makes the Matérn kernel a more realistic choice for modeling physical phenomena that are not infinitely smooth.

- **Rational Quadratic:** This kernel can be interpreted as a sum (a mixture) of RBF kernels with different length scales. This gives it the ability to model phenomena that exhibit variations on multiple scales.

$$k_{RQ}(\boldsymbol{x}, \boldsymbol{x}') = \sigma_f^2 \left(1 + \frac{\|\boldsymbol{x} - \boldsymbol{x}'\|^2}{2\alpha l^2}\right)^{-\alpha} \tag{2.6}$$

  where $\alpha > 0$ is the scale parameter. As $\alpha \to \infty$, the RQ kernel converges to the RBF kernel.

- **Periodic:** To model functions that exhibit a clear periodic trend, a specific kernel can be used:

$$k_{Per}(\boldsymbol{x}, \boldsymbol{x}') = \sigma_f^2 \exp\left(-\frac{2\sin^2(\pi\|\boldsymbol{x} - \boldsymbol{x}'\|/p)}{l^2}\right) \tag{2.7}$$

  In addition to the length scale l, the hyperparameter p appears here, defining the period of the function.

While the Squared Exponential (RBF) kernel assumes infinite smoothness of the response function, the physical behavior of structural collapse, although continuous, can exhibit rapid gradient variations near the critical limit. For this reason, the Matérn family of kernels, which allows for control over the degree of differentiability of the surrogate function, represents a more flexible and physically plausible prior choice. In particular, Matérn kernels with ν=3/2 or ν=5/2, which generate once or twice differentiable functions, are ideal candidates for modeling the relationship between geometric parameters and the collapse response.

Hyperparameter Optimization

The kernel's hyperparameters (e.g., σ_f^2, l, α, p) are not fixed a priori but are "learned" from the data. This calibration process is fundamental to adapting the model to the specific nature of the problem. The two main techniques for doing this are:

- **Maximization of the Log Marginal Likelihood:** This is the standard and most rigorous approach for Gaussian Processes. It consists of finding the set of hyperparameters that maximizes the probability (likelihood) of the observed training data. Intuitively, one seeks the kernel parameters that make the available data "most plausible." This metric, derived from the Bayesian framework, naturally balances the model's complexity (encoded in the kernel) with its fit to the data, providing inherent protection against overfitting.

- **Cross-Validation (CV):** This is a more general technique, applicable to almost all machine learning models. The training dataset is divided into k subsets (or "folds"). In turn, one of the folds is used as a validation set, while the remaining $k-1$ are used to train the model. The prediction error on the validation fold is calculated. The process is repeated k times until each fold has been used once for validation. Finally, the hyperparameters that minimize the average error calculated across all folds are chosen. Although computationally more expensive, CV can be more robust, especially if the model's assumptions (such as the Gaussian nature of the noise) are not fully met.

2.5 Comparison of Surrogate Models

Gaussian Process Regression is just one of several Machine Learning techniques applicable to the creation of surrogate models. Recent literature, such as the comparative study by Hoffer et al. [17], has analyzed the performance of different algorithms in structural mechanics contexts. The following is a critical overview of some of the most relevant alternatives, discussing their operating principles, advantages, and disadvantages in the context of this thesis.[20]

2.5.1 Multi-Layer Perceptron (MLP)

An MLP is a feed-forward artificial neural network, composed of an input layer, one or more hidden layers, and an output layer. Each neuron in a layer is connected to all neurons in the next layer via synaptic weights. The output of a neuron is calculated by applying a non-linear activation function to the weighted sum of its inputs [17]. The output of a network with L layers can be represented as a composition of functions:

$$\hat{\boldsymbol{y}}(\boldsymbol{x}) = f_L(f_{L-1}(...f_1(\boldsymbol{x})...)) \tag{2.8}$$

where the function f_l for layer l is given by $\sigma_l(\boldsymbol{W}_l^T \boldsymbol{z}_{l-1} + \boldsymbol{b}_l)$, with σ_l being the activation function, $\boldsymbol{W}_l$ the weight matrix, $\boldsymbol{b}_l$ the bias vector, and $\boldsymbol{z}_{l-1}$ the output of the previous layer.

Thanks to the universal approximation theorem, an MLP with a sufficient number of neurons can approximate any continuous function to an arbitrary degree of accuracy. MLPs have shown excellent performance in high-accuracy interpolation tasks, often outperforming other models when the dataset is sufficiently rich [17].

In the other hand their "black-box" nature makes the physical interpretation of the results difficult. They do not intrinsically provide a measure of prediction uncertainty. Their design (number of layers, neurons, choice of activation functions) is an empirical process, and training can be computationally intensive and sensitive to the choice of hyperparameters.

2.5.2 Support Vector Regression (SVR)

SVR is the regression counterpart to Support Vector Machines (SVM). The goal is not to separate classes, but to find a hyperplane that best approximates the data, minimizing the error within a certain tolerance threshold. SVR defines a "tube" of width ϵ around the regression function and tries to include as many training points as possible within it, penalizing only the points that fall outside [17]. The primal optimization problem is:

$$\min_{\boldsymbol{w},b,\xi,\xi^*} \frac{1}{2}\|\boldsymbol{w}\|^2 + C\sum_{i=1}^{n}(\xi_i + \xi_i^*) \tag{2.9}$$

subject to: $y_i - \boldsymbol{w}^T\phi(\boldsymbol{x}_i) - b \leq \epsilon + \xi_i$ and $\boldsymbol{w}^T\phi(\boldsymbol{x}_i) + b - y_i \leq \epsilon + \xi_i^*$.

The approach of SVR is effective in high-dimensional spaces, and its computational complexity does not depend on the dimensionality of the input space. By using the "kernel trick," it can efficiently model non-linear relationships.

The performance is very sensitive to the choice of the kernel and its hyperparameters (C, ϵ, γ). Like MLP, it does not natively provide an uncertainty estimate. In the study by [17], SVR showed lower performance compared to MLP and KNNR in complex structural mechanics tasks.

2.5.3 Gradient Boosting Decision Tree Regressor (GBDTR)

Gradient Boosting is an ensemble technique that builds the model sequentially. It starts with a weak model called shallow decision tree and, at each iteration, a new tree is trained to correct the residual errors of the previous model. This is done by fitting the new tree to the negative gradient of the loss function [21]. The final prediction is the sum of the predictions of all the trees:

$$\hat{y}_i = F_M(\boldsymbol{x}_i) = \sum_{m=1}^{M} h_m(\boldsymbol{x}_i) \tag{2.10}$$

This models are among the most powerful and high-performing for tabular data, often achieving very high accuracy. They are robust to outliers and do not require complex data preprocessing.

They are "black-box" models, and training can be slow. Their piecewise constant nature can be a disadvantage in approximating the continuous and smooth functions typical of physical phenomena.

2.5.4 K-Nearest Neighbors Regressor (KNNR)

KNNR is a simple non-parametric distance-based algorithm. To predict the value at a new point, the algorithm identifies the K nearest points in the training dataset (according to a distance metric, e.g., Euclidean) and calculates the mean (or a weighted average) of their output values [17].

$$\hat{y}(\boldsymbol{x}_{new}) = \frac{1}{K} \sum_{\boldsymbol{x}_i \in \mathcal{N}_K(\boldsymbol{x}_{new})} y_i \tag{2.11}$$

It is conceptually simple and easy to implement. It does not require an explicit training phase. In the study by [17], KNNR showed surprisingly competitive results, especially in extrapolation tasks and in use cases with lower geometric complexity.

The computational cost of prediction grows with the size of the dataset. It suffers from the "curse of dimensionality," and its performance degrades in high-

dimensional spaces. It does not provide an uncertainty estimate.

2.5.5 Physics-Informed Neural Networks (PINN)

PINNs are a class of neural networks that integrate knowledge of the physical laws governing a system directly into the training process. This is done by adding a term to the loss function that penalizes violations of the relevant partial differential equations (PDEs) [22]. The loss function of a PINN is typically composed of two terms:

$$\mathcal{L} = \mathcal{L}_{data} + \lambda \mathcal{L}_{physics} \tag{2.12}$$

where $\mathcal{L}_{data}$ is the error on the observed data (as in a standard MLP) and $\mathcal{L}_{physics}$ is the mean squared error of the PDE residuals, calculated over a set of collocation points in space-time.

They can be trained with sparse or incomplete datasets, as the physics component acts as a regularizer, guiding the solution towards physically consistent behaviors. In theory, they should generalize better and require less data.

Implementation is complex and requires an explicit formulation of the PDEs. Training is a challenge, as balancing $\mathcal{L}_{data}$ and $\mathcal{L}_{physics}$ (via the weight λ) is critical and non-trivial. As reported in [17], PINNs are still an active area of research, and their practical application in complex structural mechanics problems has proven difficult to optimize.

2.5.6 Observations

Several Machine Learning algorithms can be adopted as metamodels to replace high-fidelity FEM simulations. In the literature, comparative studies such as Hoffer et al. [23] show that performance depends strongly on the type of problem, the dimensionality of the parameter space, and data availability.
Gaussian Processes were chosen as the primary model for this work for the ability to provide predictions with uncertainty and a good accuracy even with relatively small datasets. After are compared with a Neural Network as MLP.

3

Development of the Computational Framework

This chapter describes the architecture of the computational framework developed to automate the entire analysis process, from the design of experiments and the generation of case studies to the training of the surrogate model. The pipeline was built using a combination of Abaqus Python scripts and Jupyter notebooks to ensure modularity, reproducibility, and flexibility.

3.1 Workflow Overview

The implemented workflow is a sequential chain of processes that transforms a predefined parameter space into a trained and validated predictive model. The flow can be summarized as follows:

1. **Generation of the Experimental Plan:** The input parameter space is sampled using a Design of Experiments (DoE) strategy to generate a finite set of configurations to be simulated.

2. **FEM Automation:** For each configuration, a Python script automatically generates the input file for the Finite Element solver (Abaqus) and manages its batch execution.

3. **Data Extraction:** The results of interest are automatically extracted from the output files of each simulation.

4. **Post-Processing:** The raw extracted data is cleaned, sorted, and consolidated into a single structured dataset. Characteristic problem parameters are also calculated.

5. **Metamodeling:** The final dataset is used to train and validate the metamodels based on Gaussian Processes, MLP and 1D predictions.

Figure 3.1 shows the workflow pipeline.

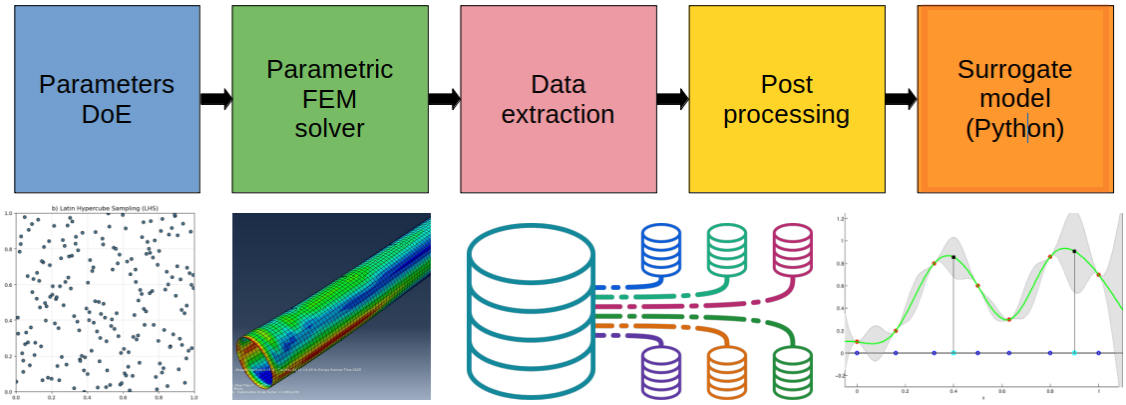


Figure 3.1: Schematic of the implemented computational workflow.

3.2 Definition of the Parameter Space

The starting point is the definition of the design space and its sampling. This operation was carried out in the notebook `00_DoE.ipynb`.

To efficiently sample the parameter space, the *Sobol Sampling* technique, discussed in section 2.3.3, was adopted. The Python library `scipy` was used to generate a matrix of N samples. The input parameters chosen to characterize the structural behavior of the Hollow Element and their respective variation ranges are summarized in Table 3.1.[25]

An equispaced grid was adopted for the main loading condition, namely the *curvature*, while for the other parameters, the low discrepancy sampling technique was used. This hybrid approach was chosen to ensure a systematic and exhaustive

exploration of the model's response across the primary loading axis (curvature), while maintaining high sampling efficiency in the multi-dimensional space of the secondary geometric and material parameters.

Table 3.1: Input parameter space for the Design of Experiments.

Parameter	Description	Range	Unit
`D_out`	Outer diameter of the Hollow Element	[32 – 42]	inches
`Wall_Thickness`	Wall thickness	[25 – 35]	mm
`Compression_Force`	Axial compression force	[1 – 5]	MN
`External_pressure`	External pressure on the Hollow Element	[0.5 – 2]	MPa
`Young Modulus`	Young's Modulus	[190000 – 230000]	MPa
`SY1`	Initial yield stress	[350 – 400]	MPa
`SY2`	Ultimate limit stress	[450 – 500]	MPa
`EP2`	Equivalent plastic strain	[0.1 – 0.2]	
`Ovalisation`	Geometrical imperfection of the section	[0 – 0.003]	
`Curvature`	Imposed curvature	[0.05 – 0.14]	rad

To concretely illustrate the procedure, the Python code block 3.1, extracted from the 00_DoE.ipynb notebook, defines the input parameter limits and performs the sampling using the Sobol sequence, generating the *Parameters_DoE.csv* file, which constitutes the list of experiments to be performed and serves as input for the next phases.

Listing 3.1: Domain definition and sample generation with the Sobol sequence method.

```
num_scen = 256  # Number of base scenarios
num_curv = 4    # Number of curvatures per scenario
curv_bounds = (0.05, 0.14)  # Range of curvatures

# Input parameters domain
bounds = {
        "D_out": (32, 42),
        "Wall_Thickness": (25, 35),
        "Compression_Force": (1e6, 5e6),
        "External_Pressure":(0.5, 2),
        "Young_Modulus":(190000, 230000),
        "SY1":(350, 400),
```

```
        "SY2":(450, 500),
        "EP2":(0.1, 0.2),
        "Ovalisation":(0, 0.02)
}

# Sampling with Sobol
sampler = qmc.Sobol(d=len(bounds), scramble=True)
samples = sampler.random(num_scen)

# Scale data in the domain edges
scaled_samples = qmc.scale(samples,
        [bounds[key][0] for key in bounds],
        [bounds[key][1] for key in bounds])

# Append curvatures to the other parameters
```

3.3 Automation of FEM Pre-processing and Solution

To avoid the manual creation of a lot of FEM models, the pre-processing and solution process was fully automated.

The core of this automation is a Python script (01_CREATE_INP.py) that iteratively reads the parameters from the CSV file generated by the DoE and builds the corresponding FEM model. The listing 3.2 shows the main function that orchestrates this process.

Listing 3.2: Main function for the automated generation of Abaqus .inp files.

```
def main():
        csv_file = "Parameters_DoE.csv"
        with open(csv_file) as file:
        reader = csv.reader(file)
        next(reader)  # Skip header
```

```
for i, row in enumerate(reader, start=1):
# ... Reading parameters from the CSV row ...
THK = float(row[1])
DIAM_p = float(row[0])
# ...

DIAM = pollici_a_mm(DIAM_p) # from inch to mm

# Functions to create the model
part_module(DIAM, OVAL)
material_module(E, NI_def, SY1, EP1_def, SY2, EP2)
section_module(THK)
assembly_module()
steps_module()
constraints_module()
amplitudes_module()
loads_module(COMPR, PRESS)
BC_module(CURV)
partitions_module()
mesh_module()
job_module(i)
```

The model was parameterized using Abaqus macros: by recording the creation of a base model, the script used for generating all input files was easily created.

Subsequently, the script `02_SOLVER.py` manages the batch execution of the simulations. It iterates over all the generated `.inp` files and submits them to the Abaqus solver via command-line calls.

To reduce the total computation time, the simulations were run in parallel. The 02_SOLVER.py script manages a process pool, launching a predefined maximum number of simultaneous Abaqus analyses, as shown in the code snippet 3.3.

Listing 3.3: Parallel process management for running Abaqus simulations.

```
MAX_PARALLEL_JOBS = 2
```

```
running_processes = []
job_index = 0

while job_index < len(jobs_to_run):
  if len(running_processes) < MAX_PARALLEL_JOBS:
    job_name = jobs_to_run[job_index]
    input_path = os.path.join(input_folder, job_name + ".inp")
    job_output_folder = os.path.join(output_folder, job_name)
    # ...
    command = "abaqus␣job={}␣input={}"
        .format(job_name, absolute_input_path)
    process = subprocess
        .Popen(command, shell=True, cwd=job_output_folder)
    running_processes.append(process)
    job_index += 1
  else:
    # Wait for a process to finish before starting another
    time.sleep(15)
```

The limitations can be the number of processors on the execution machine, the number of concurrently available Abaqus licenses, and the amount of computer RAM. The latter was the bottleneck encountered in this project. The FEA simulations were performed on a machine from the *PINECA_Lab_P200_Magistrale*.

If the bottleneck were the number of licenses, another feasible but less efficient path is the multiprocessor parallelization of Abaqus on a single *Job*.

3.4 FEM Model: PIPE_1

3.4.1 Model description

The case study consists of a thin-walled metal Hollow Element, identified as `PIPE-1`, discretized with `S4R` quadrilateral shell elements. These 4-node elements with reduced integration are suitable for nonlinear analyses involving large rotations and buckling phenomena [26, 1]. The mesh size is discussed below.

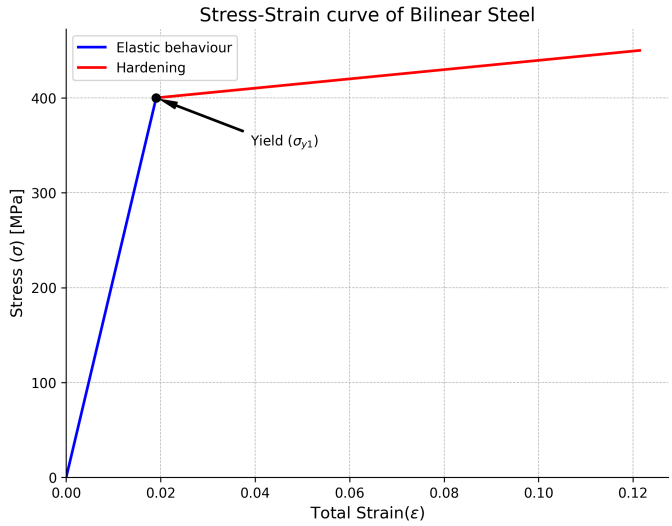


Figure 3.2: 1D representation of the Stress-Strain curve

The material is steel, modeled with an elasto-plastic behavior. The elastic response is assumed to be linear, isotropic, and defined by Young's Modulus and Poisson's ratio. The post-yield phase is described by a J_2 plasticity law (von Mises) with bilinear isotropic hardening, defined by two points on the plastic stress-strain curve, as in Figure 3.2.

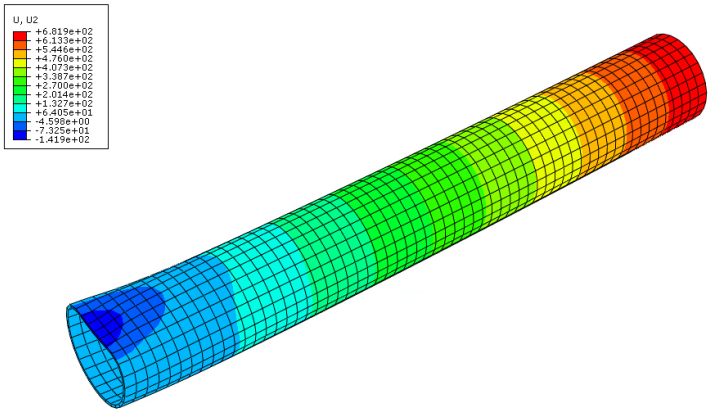
This approach allows for capturing both the initial elastic response and the ductile collapse typical of metals [4, 27].

3.4.2 Mesh size

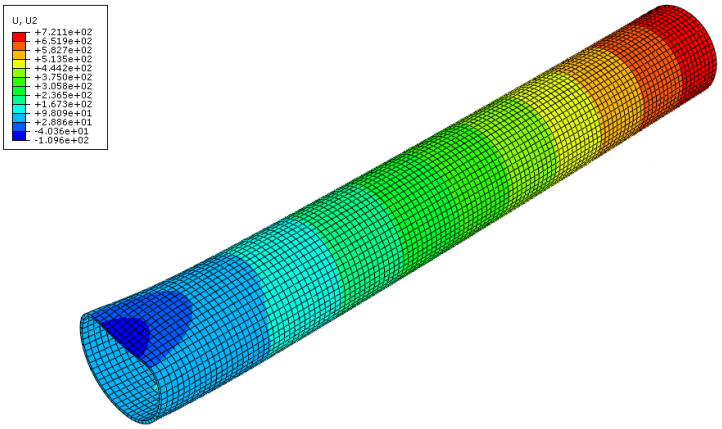
The choice of spatial discretization represents a critical compromise between solution accuracy and computational cost. A preliminary analysis was conducted to evaluate the influence of mesh density on the results.

As qualitatively illustrated in Figures 3.3a, 3.3b, and 3.3c, different discretizations were compared.

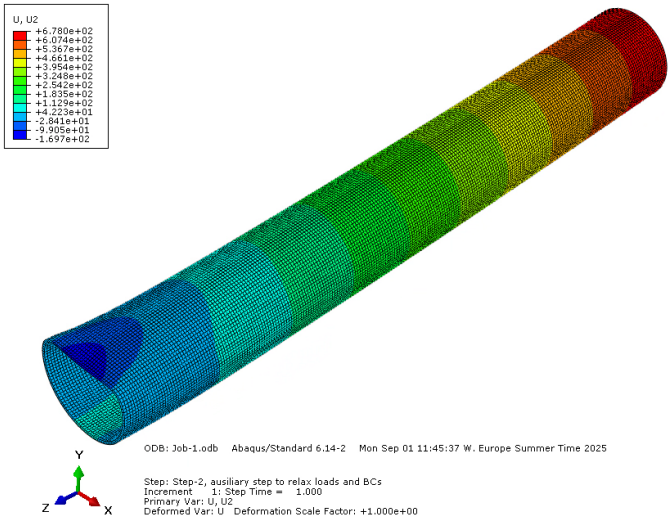
An analysis was performed with the following parameters: curvature 0.14 [rad], 4 processors, Axial compression 2 [MN], thickness 25 [mm], outer diameter [40 inches].



(a) Y deformation (U2) with the coarsest mesh (100mm)



(b) Y deformation (U2) with the middle mesh (50mm)



(c) Y deformation (U2) with the finest mesh (25mm)

Table 3.2: Analysis results

Mesh size	n nodes	n elements	Wall clock time (s)	AR (%)	DR (%)
100 mm	1613	1586	24	1.33108	19.23202
50 mm	6643	6588	83	7.07742	24.68337
25 mm	26461	26352	326	7.07278	23.56646
18 mm	51001	50850	736	7.06719	24.01805

Data analysis in Table 3.2 shows that the mesh with 100 mm elements is clearly insufficient. The output metrics (AR and DR) diverge significantly from those of finer discretizations, indicating that this mesh cannot correctly capture the physics of the problem.

Switching to the 50 mm mesh stabilizes the solution. A considerable convergence towards the final values is observed, especially for the AR (Area Reduction) metric, which increases from before.

Further refinements (from 50 mm to 25 mm and then to 18 mm) show that the solution has reached convergence. The AR metric asymptotically stabilizes around the value of 7.07%. The DR (Diameter Reduction) metric, despite showing a slight oscillation, stabilizes within a narrow range ($\sim$23.5% - 24.7%), with marginal variations between the last refinement steps.

Based on these results, the 50 mm mesh was chosen for the simulation campaign. This decision is based on a cost-benefit analysis: the switch from 100 mm to 50 mm produces a fundamental improvement in accuracy for an acceptable increase in computation time (from 24 to 83 seconds). Subsequent refinements, while marginally improving precision, lead to an order-of-magnitude increase in computational cost (326 and 736 seconds).

Therefore, the 50 mm mesh represents the optimal choice that ensures an adequate level of accuracy for the purposes of this surrogate model, while keeping the computational burden within manageable limits. For real world application could be better to use a finer mesh grid, for robustness and stability in all the possible configurations.

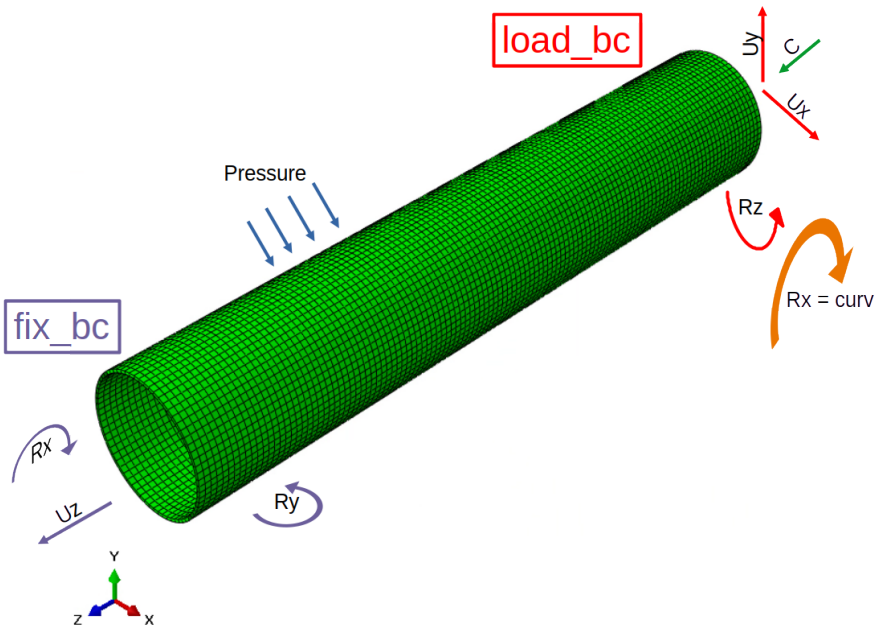


Figure 3.4: Boundary conditions and loads application

3.4.3 Boundary conditions and loads

Two main constraints were applied at the ends of the Hollow Element:

- `fix_BC`: degrees of freedom ($U3 = UR1 = UR2 = 0$) were fixed at one end;
- `load_BC`: a constraint at the other end ($U1 = U2 = UR3 = 0$) that allows the application of the compression force and the imposed curvature.
- *curvature_BC*: an imposed curvature ($UR1 = CURV$) is forced on the *load_BC* end, which imposes a rotation to simulate the bending of the Hollow Element, as in figure 3.4.

The applied loads, representative of the stresses acting on a Hollow Element, are applied at *load_BC* end, and on the external surface of the entire body, respectively:

- `compression_load`: a concentrated axial compression force;
- `ext_pressure`: a uniform pressure applied to the outer surface of the Hollow Element.

3.4.4 Analysis strategy

The analysis was conducted in two sequential steps to simulate the loading and subsequent unloading of the structure:

- **Step-1 (Loading)**: Simultaneous and proportional application of the compression force, external pressure (amplitude from 0 to 0.5), and subsequently the imposed curvature (amplitude from 0.5 to 1). The analysis continues until the full load is reached, capturing the structural response up to potential collapse. *(Static, General)*
- **Step-2 (Unloading)**: Complete and simultaneous removal of all loads and the curvature constraint applied in Step-1. This step allows for the calculation of the residual deformation state and the elastic spring-back of the structure. *(Static, General)*

The problem is highly nonlinear, due to geometric nonlinearity (large displacements and local/global instability, `nlgeom=ON`) and material nonlinearity (plasticity of the steel).

The solution is obtained using an implicit static solver that employs an incremental Newton–Raphson method with automatic step size control. To improve convergence and follow the equilibrium path even in the post-buckling phase (i.e., after the onset of instability), an automatic stabilization technique based on energy dissipation was activated. This approach adds fictitious damping to the model, allowing the solver to overcome bifurcation or limit points (instability) without resorting to more complex methods like the arc-length method (Riks).
This latter method would have been numerically more consistent, but in the available Abaqus licence it is not possible to perform an unloading step after it. Therefore, it's not usable for analyzing the unloaded deformation, but rather for a collapse analysis.

3.5 Data Extraction and Post-processing

Once the simulations are complete, the data of interest (nodal coordinates and displacements) are extracted from the binary output files (.odb) using a Python script (03_DATA_EXTR.py). Afterwards, the data were grouped by sections along Z

coordinate and sorted by counterclock-wise (04_SORTING.ipynb). Subsequently, a notebook (05_POST_PROC.ipynb) calculates the synthetic collapse indicators, which are essential for training the surrogate models.

The output parameters of interest, calculated for the surrogate model, are:

- Area Reduction (AR): calculates $\frac{AN-AD_{min}}{AN} \cdot 100$, where AN is the nominal area and AD_{min} is the minimum deformed area. *(Worst-case parameter)*
- Diameter Reduction (DR): calculates $\frac{DN-DD_{min}}{DN} \cdot 100$, where DN is the nominal diameter and DD_{min} is the minimum deformed "vertical" diameter. *(Worst-case parameter)*
- Area Profile (AP): calculates the deformed area for each curvilinear abscissa, along the length of the pipe axis Z: it is the only *vectorial parameter*.

Listing 3.4: Routine for calculating AR and DR indicators from extracted data.

```
# Calculate nominal area from the first section (Z_min)
A_nom =
    df_filtered[df_filtered['Z'] == Z_min]['Area'].iloc[0]
D_nom_vert =
    df_filtered[df_filtered['Z'] == Z_min]['Diam_Vert'].iloc[0]

# Find the minimum area and minimum vertical diameter
A_min = df_filtered['Area'].min()
D_min_vert = df_filtered['Diam_Vert'].min()

# Calculate AR and DR as percentages
AR = ((A_nom - A_min) / A_nom) * 100
DR_vert = ((D_nom_vert - D_min_vert) / D_nom_vert) * 100

# Create summary
summary_df = pd.DataFrame({
            'AR␣(%)': [AR], 'DR␣(%)': [DR_vert]
            })
```

For robustness, it was decided to use the last available increment of *Step-1* if the analysis could not proceed to the end of *Step-2*. This situation has occurred about 1% of the time.

The final output of this stage is the file `AR_DR_results.csv`, a clean and structured dataset that forms the knowledge base for training the metamodel. For each run, a file with the deformed and undeformed area and diameter profiles, and their respective plots, are also generated. [28]

3.6 Metamodeling

As the final most-relevant stage of this computational workflow, the clean and structured dataset *AR_DR_results.csv* is used to train and validate the surrogate models. This metamodeling phase involved the development of two distinct surrogate models, or metamodels, to approximate the complex input-output relationship of the simulated mechanical system. The objective is to create computationally inexpensive predictors for key performance indicators, namely Diameter Reduction (DR), Area Reduction (AR) and an indicative behaviour of Area Profile (AP).
Two paradigms from machine learning were investigated: a probabilistic, non-parametric approach using Gaussian Process Regression (GPR), and a parametric approach using a Multi-Layer Perceptron (MLP). Both models were implemented in a Python environment, leveraging specialized libraries for robust execution.

3.6.1 Gaussian Process Regression (GPR) Metamodel

The GPR model offers a Bayesian perspective on regression, defining a probability distribution over a space of functions. This inherent probabilistic nature allows for principled uncertainty quantification. The implementation relies on the `GaussianProcessRegressor` from the `scikit-learn` library.

The first operational step was, naturally, a rigorous pre-processing of the dataset sourced from `AR_DR_results.csv`. The input feature space, $\mathbf{X} \in \mathbb{R}^{n \times d}$, was subjected to a standardization procedure using the `StandardScaler`. This transformation re-scales each feature to have a zero mean and unit variance, which is crucial

for kernel-based methods as it ensures isotropic assumptions are valid and improves the numerical conditioning of the optimization problem.
The target variable, Diameter Reduction (y_{DR}), underwent a logarithmic transformation, $y'_{DR} = \log(y_{DR}+1)$, to stabilize its variance and map the positively skewed, non-negative response to a more Gaussian-like distribution, better aligning with the model's underlying assumptions.

The core of a GPR model is its kernel, or covariance function, $k(\mathbf{x}_i, \mathbf{x}_j)$, which encodes the prior belief about the function's properties. A composite kernel was formulated to capture the heterogeneous nature of the system's response. This kernel is a product of a `ConstantKernel`, which scales the magnitude of the covariance, and a sum of two distinct kernels: a Radial Basis Function (RBF) kernel and a Matérn kernel with $\nu = 3/2$. The RBF kernel, $k_{RBF}(\mathbf{x}_i, \mathbf{x}_j) = \sigma^2 \exp\left(-\frac{1}{2l^2}||\mathbf{x}_i - \mathbf{x}_j||^2\right)$, assumes infinite differentiability and is thus well-suited for modeling smooth, global trends. The Matérn kernel is less smooth (once differentiable for $\nu = 3/2$) and offers greater flexibility in modeling more complex, local phenomena with higher frequency variations.

The model's training phase consisted of optimizing the kernel's hyperparameters. This was performed using the *L-BFGS-B* optimization algorithm, a quasi-Newton method suitable for this unconstrained problem. To mitigate the risk of convergence to sub-optimal local minima in the LML landscape, a multi-start optimization strategy was employed with 40 restarts from different initializations (`n_restarts_optimizer=40`).

For validation, a robust scenario-level Leave-One-Out Cross-Validation (LOOCV) protocol was implemented. This involves iteratively holding out a full operational scenario as the test set while training the model on the remaining data. This technique provides an unbiased estimate of the model's generalization performance on unseen conditions. The model's predictive accuracy was quantified using the classical coefficient of determination (R^2), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE).

3.6.2 Multi-Layer Perceptron (MLP) Metamodel

As an alternative, a Multi-Layer Perceptron was developed using the `PyTorch` deep learning framework. This parametric model approximates the target function through a series of nested, non-linear transformations learned from the data. Data handling for the MLP followed established deep learning practices. The same source dataset was utilized, and for the prediction of Area Reduction (AR), both the input features and the target variable were standardized. The data was partitioned at the scenario level into training, validation, and test sets. This ensures that the model is evaluated on its ability to generalize to entirely new loading curves, preventing data leakage. The datasets were managed by the `DataLoader` class, which facilitates efficient, batched processing during training.

The network architecture was defined as a sequential model with two hidden layers. The input layer accepts a tensor of 10 features. The first hidden layer consists of 64 neurons, and the second contains 32 neurons. Both hidden layers employ the Rectified Linear Unit (ReLU) as the activation function, $f(x) = \max(0, x)$, to introduce non-linearity. The final output layer is a single linear neuron that produces the scalar prediction.

The network's weights were optimized by minimizing the Mean Squared Error (MSE) loss function using the Adam optimizer, an adaptive learning rate optimization algorithm. The learning rate was set to 10^{-3} and 2×10^{-3} for the DR and AR models, respectively. The training was conducted for a set number of epochs (200 for DR, 100 for AR). To prevent overfitting, an early stopping mechanism was implemented. This technique involves monitoring the loss on the validation set at the end of each epoch and saving the model state corresponding to the minimum validation loss. This ensures that the selected model is the one that exhibits the best generalization performance on data not seen during training.
The final evaluation of the trained network was performed on the held-out test set, using the model weights saved during the early stopping procedure. The performance was assessed using the same suite of metrics (R^2, RMSE, MAE) as the GPR model to enable a direct and impartial comparison of the two distinct metamodeling approaches.

4

Preliminary Analysis: 1D Surrogate Models for Curvature Interpolation

4.1 Objective of the Preliminary Analysis

The idea of studying a single scenario stems from an initial attempt to speed up the simulation of the phenomenon under analysis. The objective is to predict the behavior of the Hollow Element subjected to various imposed curvatures, knowing only a few of them derived from the high-fidelity FEM model analysis. This section will illustrate a preliminary study conducted on a dataset produced by a preliminary model, which was considered adequate for understanding the framework, as it models the same phenomenon.

4.1.1 Database description

In this part, we will focus on the analysis of a single scenario, to create a model that predicts all the curves showing the parameters interest and curvature, given only a few known training curves.

Initially, the plan was to create a model that considers 3 to 5 known curves and predicts the others within the scenario (an additional 13-15 curves).

The prediction parameters of interest are the area profile vector (AP), the scalar parameter of maximum area reduction (AR), and the scalar one of outer diameter reduction (DR).
The formulas to derive the two scalar parameters have been mentioned before.

4.1.2 Description of the model used

For the prediction of the parameters described above, simple interpolation and Machine Learning models were analyzed and then compared. Given the low availability of data for each scenario, the complexity of the model class used was limited to avoid overfitting.

The models compared were:

- Polynomial regression of degree 2
- Natural cubic spline
- Piecewise Cubic Hermite Interpolating Polynomial (PCHIP)
- Gaussian Process with a *rationalquadratic* kernel
- Gaussian Process with a *matern32* kernel

The polynomial regression used finds the coefficients using the least squares method.

The Spline interpolation divides the data interval into multiple subintervals and constructs a third-degree polynomial for each. These polynomials are chosen so that they pass through all data points and are continuous up to the second derivative, ensuring a smooth transition.

The Piecewise Cubic Hermite is a regression model similar to the Spline but guarantees continuity only of the first derivative, maintains the monotonicity of the data, and is more robust against unwanted oscillations.

The Gaussian Process is a non-parametric probabilistic model described by a mean function and a kernel, which assigns a Gaussian distribution to every set of points. In this case, two Matlab kernels were compared to understand how this choice affects the model's performance.

4.1.3 Leave-One-Out (LOO) validation

The metric to establish the performance of the various models was based on Leave-One-Out (LOO) validation, which involves training on all available sample curves except one, and then comparing the model's prediction with the actual known value of the excluded curve.

Previous work has shown that this validation provides an upper bound of the error, proving this approach effective in evaluating model performance with limited data. We take this as a starting hypothesis.

For the DR and AR parameters, the mean of the relative percentage errors was then calculated, while for the area profile, the mean of the relative percentage errors, also averaged over the size of the curvilinear abscissa, was used.

This metric later proved inadequate for comparison with the AR and DR errors, and it was decided to also include the maximum relative percentage error of the area profile, which aligns better with the values of the other two errors.

4.1.4 Model Comparison

A battery of 5 datasets with an increasing number of available curves was prepared, on which the models were trained and compared using the parameters derived from LOO validation. The results of this comparison are summarized in Table 4.1.

It can be noted that polynomial regression does not provide a sufficiently accurate prediction in any test case. Spline and PCHIP are the most accurate with a small amount of data, while Gaussian Processes perform similarly but only begin to align with the performance of polynomials with a high number of input curves. Given the specific problem requirement of having an effective prediction with little data, Gaussian processes were excluded for now.

4.1.5 Domain extensions

A feature asked in the model was domain extension, in order to also provide a description of an area outside the available points.

The addition of the origin point (0,0), signifying an absence of deformation in the absence of external load, helped to significantly improve the model's performance. This point derives from a physical consideration and also improved the prediction of the domain extension towards the lower load conditions.

The opposite extension of the domain proved more complex. The only two models that seemed to perceive a more faithful global trend were polynomial and spline regression, while the others, in the majority of cases examined, showed poor prediction outside the analysis domain.

4.1.6 Hybrid model

To try to achieve the best result in terms of both local and global trends, it was decided to create a hybrid model, using a Piecewise Cubic Hermite regression for the internal values and the left extension, and a spline or polynomial regression for the right extension. Both regressions were trained on all available points.

To blend the two models more continuously, a shape function was used with a parameter to manage the width of the transition area, as illustrated in Figure 4.1.

4.1.7 Visual Analysis of the Hybrid Model

The following figures provide a visual representation of the hybrid model's performance on a sample scenario. In Figures 4.2 and 4.3, which show the PCHIP-Polynomial model, it is possible to observe how the prediction (magenta line) accurately follows the known data points (blue dots) within the interpolation domain, thanks to the PCHIP component. The extrapolation to the right, managed by the polynomial, shows a physically plausible collapse trend. A similar behavior can be seen for the PCHIP-Spline model in Figures 4.4 and 4.5, demonstrating the robustness of this hybrid approach.

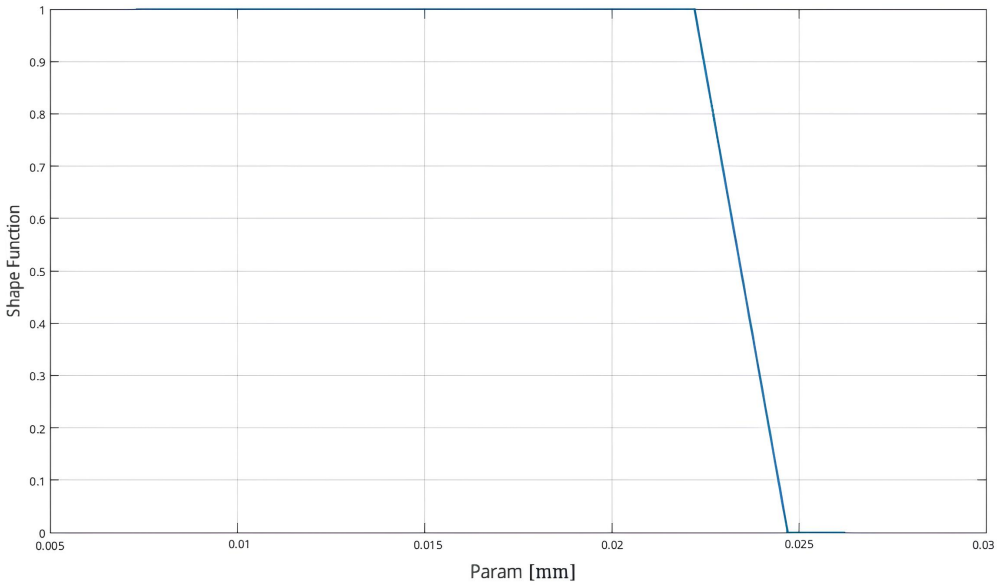


Figure 4.1: Shape function used to blend the PCHIP and polynomial/spline models.

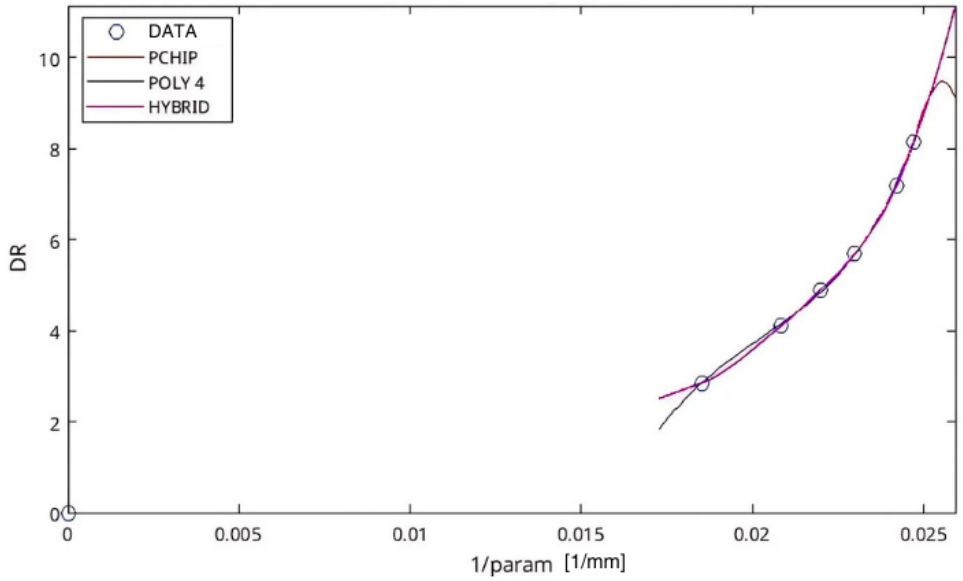


Figure 4.2: Prediction of Diameter Reduction (DR) using the hybrid PCHIP-Polynomial model.

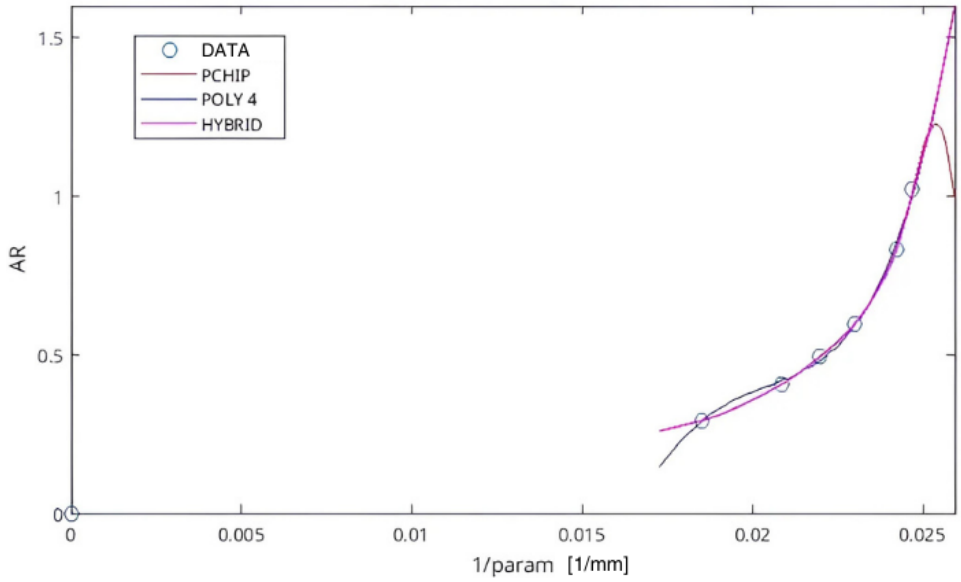


Figure 4.3: Prediction of Area Reduction (AR) using the hybrid PCHIP-Polynomial model.

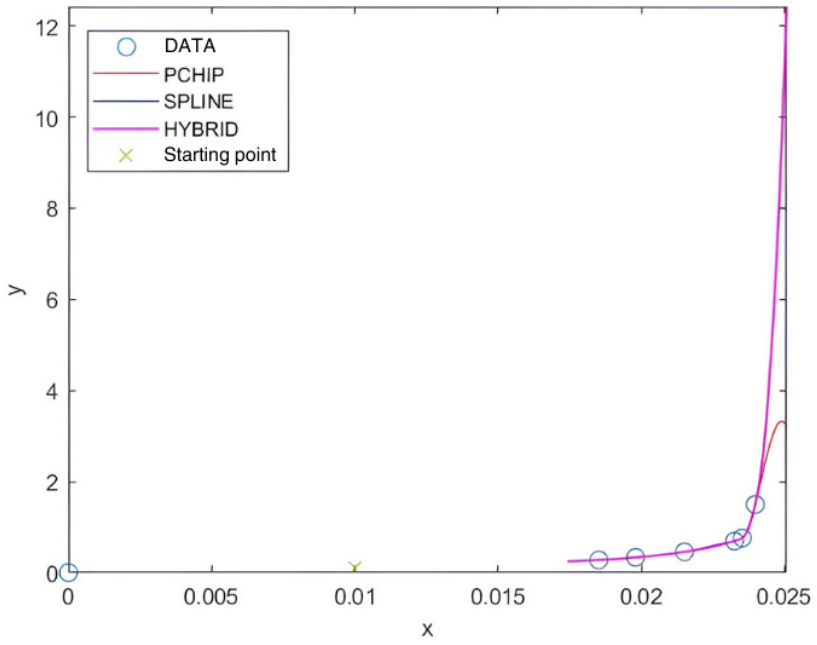


Figure 4.4: Prediction of Area Reduction (AR) using the hybrid PCHIP-Spline model.

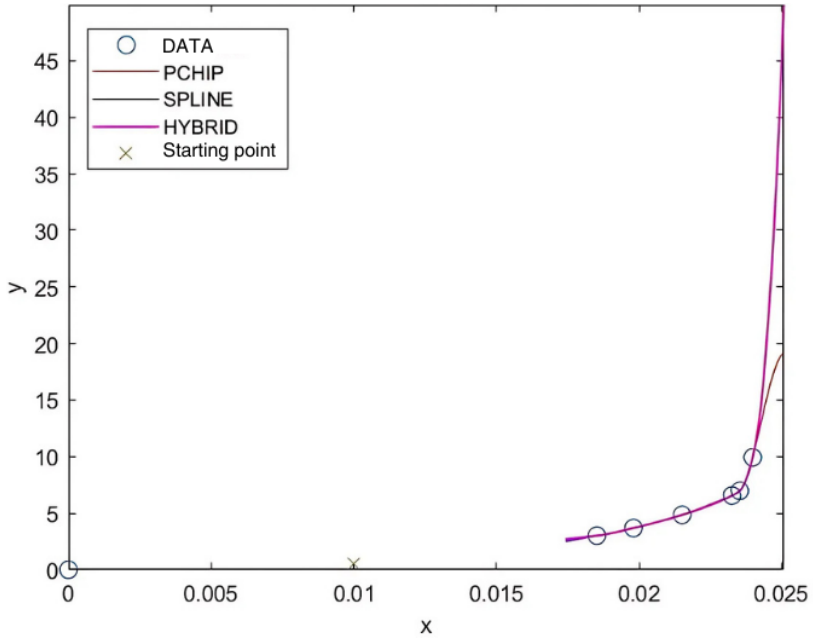


Figure 4.5: Prediction of Diameter Reduction (DR) using the hybrid PCHIP-Spline model.

4.1.8 Discussion and Limitations of the 1D Approach

The preliminary analysis in this chapter served to validate the feasibility of a surrogate model approach within an extremely limited data regime. Our comparison, based on a Leave-One-Out (LOO) metric, allowed us to quantify the performance of different models and revealed several clear trends.

Our analysis showed that the Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) emerged as the best-performing model for *intra domain* interpolation, especially with a small number of training points. Its ability to preserve the monotonicity of the training data proved crucial in avoiding the physically unrealistic oscillations we observed in higher-degree polynomial models. Furthermore, we found that introducing physical knowledge into the model, by imposing the origin point of zero curvature and zero deformation, was a fundamental step. This not only acted as an effective regularization but also ensured a physically consistent behavior near zerobload conditions, significantly improving its extrapolation capabilities toward lower curvatures. Our final solution, a hybrid model that combines PCHIP local accuracy with a polynomial's ability to capture a global trend for extrapolation, represents an effective engineering compromise.

Despite these encouraging results, the one-dimensional (1D) approach presented here suffers from fundamental limitations that preclude its use as a general predictive tool.

- Lack of Generalizability: The model is *scenario specific*. We trained it to interpolate the effect of curvature alone while all other physical parameters were held constant. As a result, the surrogate does not learn the underlying physics but only the functional relationship for a single case. Any variation in other parameters would require a new, expensive set of FEM simulations and a complete retraining of the model.

- Oversimplification of the Parameter Space: The real problem is multi-dimensional. By reducing the analysis to only curvature as the independent variable, we created a drastic simplification that ignores the coupling effects between different physical parameters. Consequently, the model has no predictive capability outside the narrow path sampled in the parameter space.

- Heuristic Nature of Extrapolation: Our hybrid model, although functional, is based on a heuristic. The choice of the blending function and the model for extrapolation were not derived from first principles but from an a posteriori observation of the data. The reliability of this extrapolation is therefore not guaranteed and remains an unknown, especially for curvatures significantly outside the training range.

In conclusion, this 1D analysis was a valuable and successful exercise. We demonstrated that it is possible to build an accurate surrogate for interpolation within a single scenario with very few data points. We also identified PCHIP as the most suitable model class and highlighted the importance of integrating physical constraints. However, the intrinsic limitations, particularly the lack of generalizability, make this approach inadequate for our final goal of creating a robust predictive tool. This preliminary study therefore provides a clear justification for moving beyond the one-dimensional paradigm toward multi-dimensional surrogate models, which can map the entire input parameter space to the outputs of interest. Nonetheless, this 1D approach remains a viable solution for specific cases where configurations are stable and only the imposed curvature varies, offering a good prediction of the structure's collapse.

4.2 Results on the case study model

The operational procedures studied on the preliminary model were then applied to the case study model. The results are in line with expectations.

Initially, using the Leave-One-Out and Grid-Search approach, the following hyperparameters were tuned:

- Degree of the polynomial for global interpolation:
  $degrees = [2, 3, 4]$.
- Defines the base weight, to always mix the two models slightly:
  $mix_mod_values = [0.65, 0.75, 0.85, 0.9]$.
- Controls the width of the region where the transition between the two models occurs: $mix_par_values = [0.4, 0.55, 0.7, 0.8]$.

The best combinations were found for *Diameter Reduction* with $(2, 0.75, 0.7)$ and a Mean Square Error of approximately 27.33. For the *Area Reduction* parameter, the values $(2, 0.9, 0.4)$ were found, which produce a model with an MSE of about 14.67.

Analyzing Figures 4.6 and 4.7, it can be observed that in this case as well, the choice of the hybrid model prevents non-physical behaviors, creating a very reliable prediction in the curvature domain $[0, 0.14]$ and an indicative one for higher values. The origin point, which implies no deformation in the case of zero curvature, added based on physical heuristics, proves to stabilize the model in this circumstance as well.

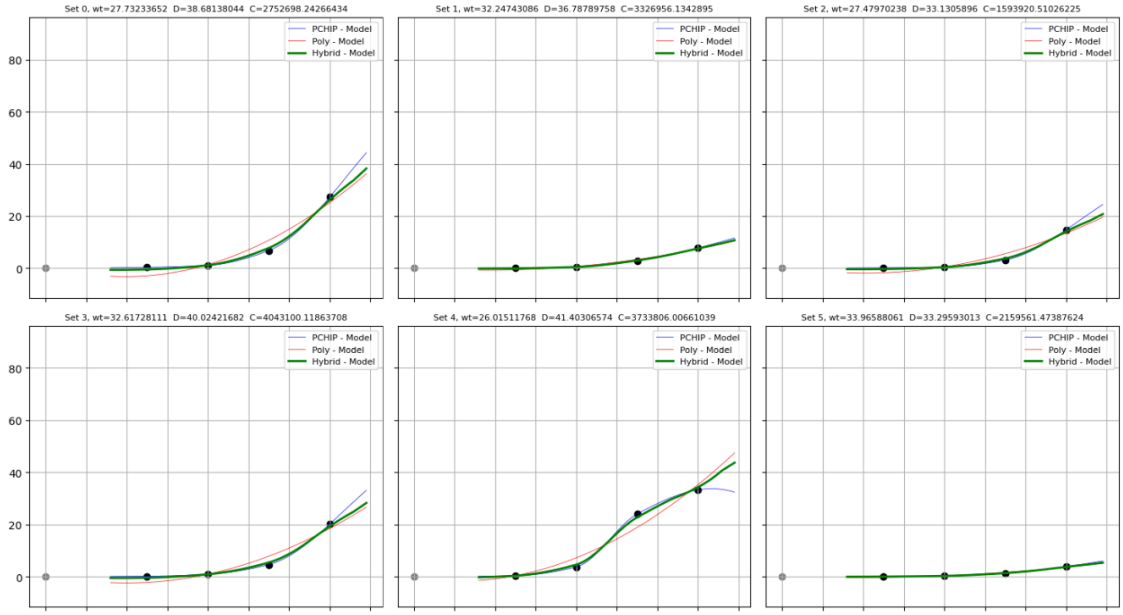


Figure 4.6: Diameter reduction prediction for some of the scenarios

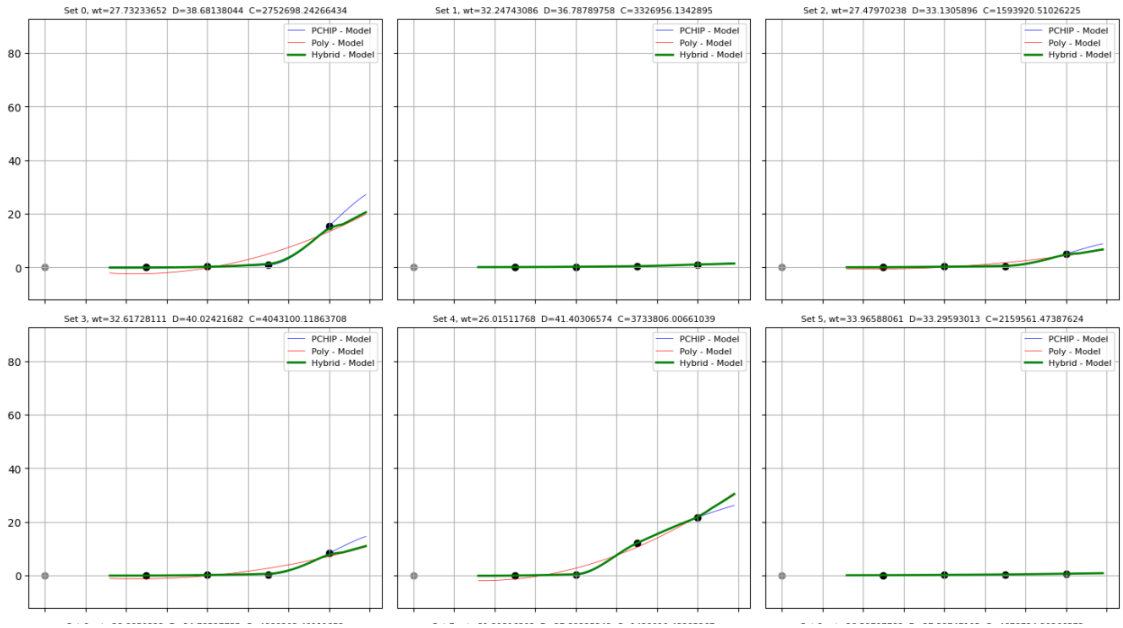


Figure 4.7: Area reduction prediction for some of the scenarios

	AP	AR	DR
Train set 1 (3 points)			
Model 1: Poly deg 2	0.04135	538.5264	662.7563
Model 2: Spline	0.041335	646.1013	489.2474
Model 3: PCHIP	**0.023768**	**81.2759**	**54.9047**
Model 4: GP rat quadratic	0.03089	261.5039	344.4467
Model 5: GP matern 32	0.032576	207.9136	325.4226
Train set 2 (5 points)			
Model 1: Poly deg 2	0.019	321.754	389.4538
Model 2: Spline	0.0085162	152.2609	117.1716
Model 3: PCHIP	**0.0081018**	**23.6728**	**27.5434**
Model 4: GP rat quadratic	0.015789	114.5669	151.3395
Model 5: GP matern 32	0.015514	128.5927	200.3511
Train set 3 (8 points)			
Model 1: Poly deg 2	0.010572	186.497	221.5355
Model 2: Spline	0.0032245	27.2102	**12.9118**
Model 3: PCHIP	**0.0030159**	**8.5412**	13.3674
Model 4: GP rat quadratic	0.0067638	55.0097	81.9762
Model 5: GP matern 32	0.0063503	53.0079	63.5042
Train set 4 (11 points)			
Model 1: Poly deg 2	0.0076509	144.5092	164.3697
Model 2: Spline	0.0038387	**8.6767**	37.2671
Model 3: PCHIP	**0.0022778**	6.0449	**9.3999**
Model 4: GP rat quadratic	0.095784	37.574	60.195
Model 5: GP matern 32	0.0044275	41.2822	62.1558
Train set full (16 points)			
Model 1: Poly deg 2	0.0053457	100.0034	120.2355
Model 2: Spline	0.0035802	12.9659	55.32
Model 3: PCHIP	**0.001908**	**3.1753**	6.1724
Model 4: GP rat quadratic	**0.0018421**	22.0444	31.7013
Model 5: GP matern 32	0.0019341	9.8375	**4.7002**

Table 4.1: Mean relative percentage error (%) from LOO validation for different models and training set sizes. High values indicate poor predictive performance, especially with few data points.

5

Analysis and Results of the Multivariate Surrogate Model

This chapter presents the final results of the computational framework. The Gaussian Process Regression (GPR) metamodel and a Multi-Layer Perceptron (MLP) approach, trained on this dataset, are quantitatively and qualitatively evaluated to verify their predictive capabilities.

5.1 Simulation Results

The training dataset was generated by executing a set of nonlinear Finite Element Method (FEM) simulations in Abaqus. The configurations for these simulations were determined by an experimental plan created using the Sobol sampling technique, a low discrepancy sequence method chosen to ensure uniform coverage of the parameter space, as detailed in Chapter 3.

5.1.1 Extracted Area Profiles

Figures 5.1 and 5.2 show two representative examples of the simulation output. The plots compare the cross-sectional area profile of the Hollow Element along its axial coordinate between the nominal configuration and the final deformed configuration

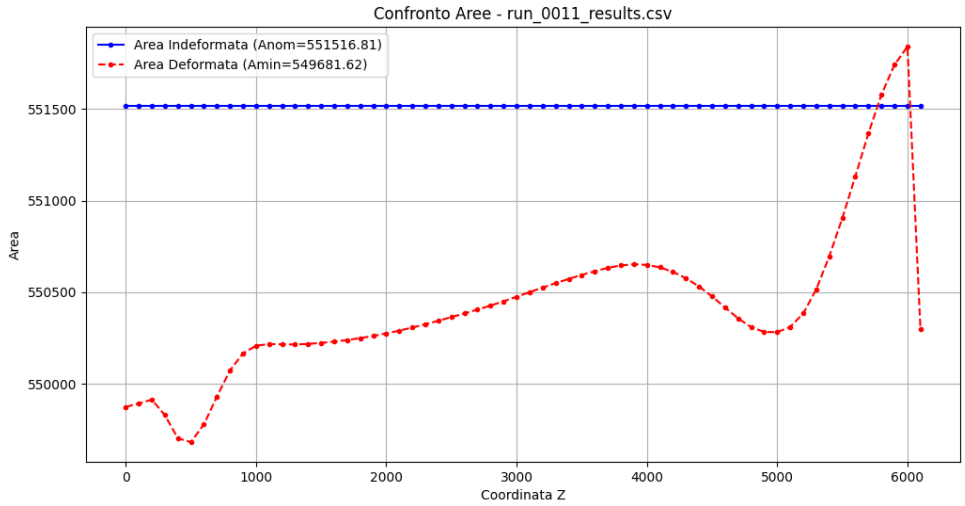


Figure 5.1: Comparison between the nominal area profile (blue line) and the deformed area profile after unloading (red line) for an imposed curvature of 0.11. The section reduction is limited and distributed, with no localized collapse.

(post-unloading). These visualizations confirm the highly nonlinear nature of the collapse phenomenon, where a relatively modest increase in the imposed curvature can trigger a catastrophic and localized structural response. From these vector profiles, the scalar metrics of interest, Diameter Reduction (DR) and Area Reduction (AR), are extracted to summarize the extent of the collapse.

5.1.2 Correlation between variables

For a preliminary quantitative assessment of the relationships between input parameters and outputs, the Pearson correlation matrix was calculated, as shown in Figure 5.3. The analysis of the matrix highlights the following physically consistent trends:

- Strong Positive Correlation: The `Curvature` shows the strongest positive correlation with the collapse metrics DR (+0.57) and AR (+0.53), as it is the main loading agent inducing instability.
- Negative Correlation: The wall thickness (`Wall_Thickness`) shows a significant negative correlation (-0.32 for both DR and AR), as an increase in thickness increases the Hollow Element's bending stiffness.

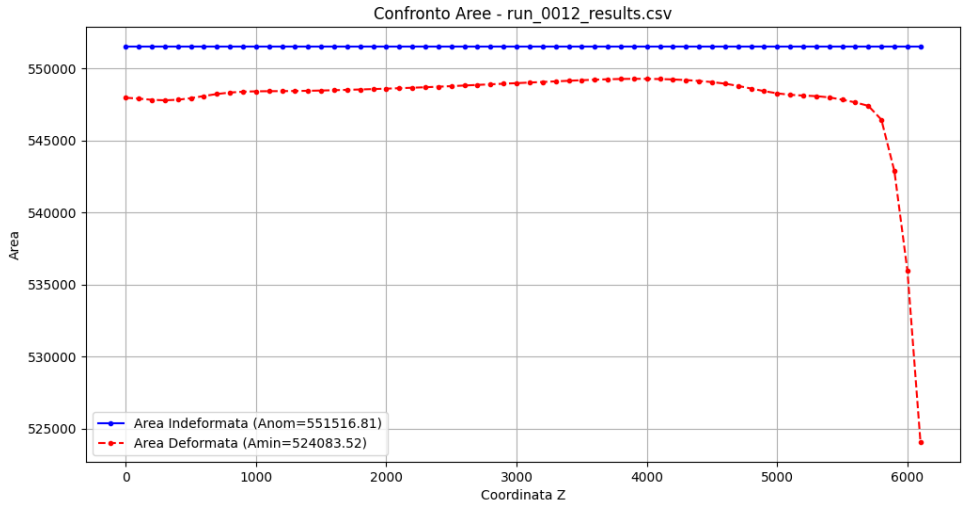


Figure 5.2: Comparison between the nominal area profile (blue line) and the deformed area profile after unloading (red line) for an imposed curvature of 0.14. A clear localized collapse is observed, with a drastic reduction of the section in a specific zone.

- Correlation between Outputs: The DR and AR metrics are almost perfectly correlated (+0.96), indicating that they essentially describe the same physical phenomenon. For subsequent analyses, we can focus on predicting only one of the two.

It is important to emphasize that this analysis only captures linear dependencies and cannot describe the complex interactions between parameters, a task left to the nonlinear surrogate model.

5.2 GPR Metamodel

The processed dataset was used to train a metamodel based on Gaussian Process Regression (GPR). As described in Chapter 3, a composite kernel (Matérn and RBF) was primarily chosen to adequately model the mixed nature of the inputs.

The GPR metamodel was implemented using the scikit-learn library. A composite kernel was defined, combining an RBF kernel and a Matérn kernel to achieve greater flexibility. The following code 5.1, from 07_PRED_GPR.ipynb, shows an example of the model's initialization and training.

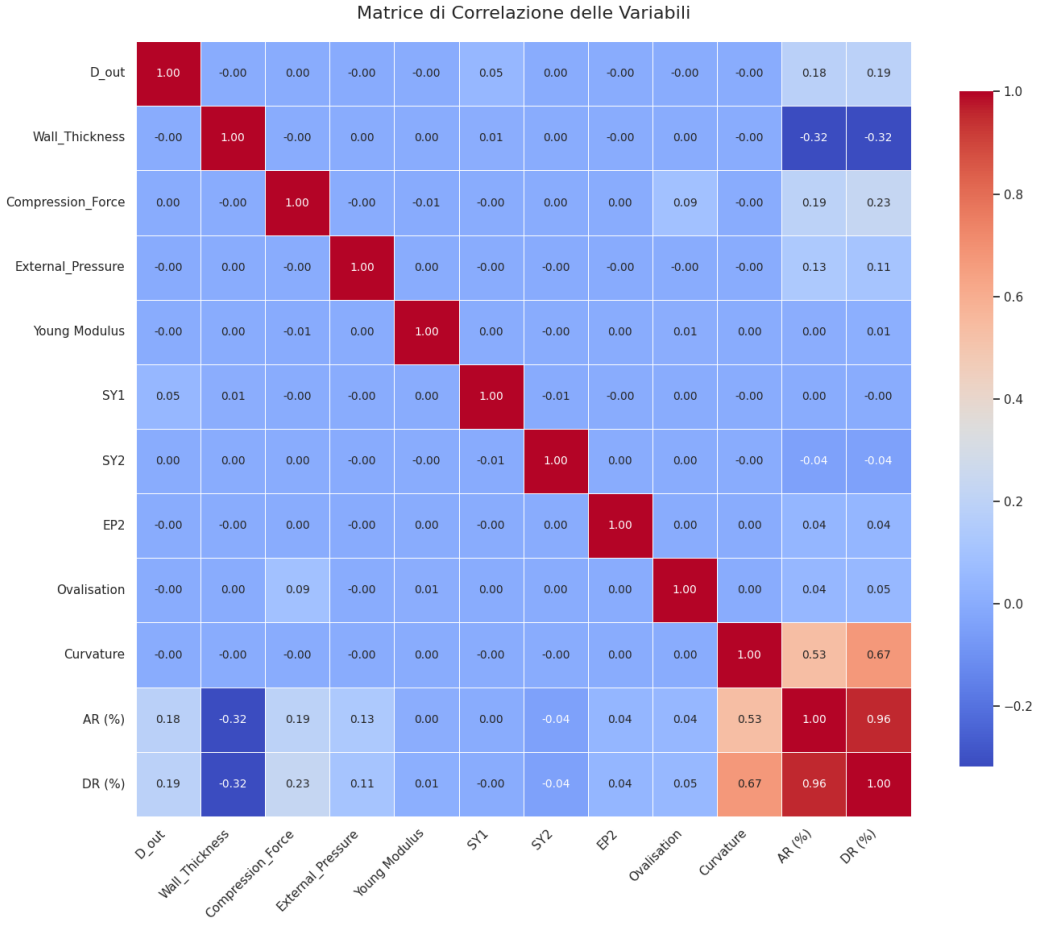


Figure 5.3: Pearson correlation matrix between the input parameters and the output metrics (AR and DR). Values close to $+1$ (red) indicate a strong positive correlation, while values close to -1 (blue) indicate a strong negative correlation.

Listing 5.1: Example of Kernel definition and use in GPR model training.

```
from sklearn.gaussian_process
  import GaussianProcessRegressor
from sklearn.gaussian_process.kernels
  import RBF, Matern, ConstantKernel

# Composite kernel for greater flexibility
kernel_1 = RBF(length_scale_bounds=(1e-05, 1e6))
kernel_2 = Matern(length_scale_bounds=(1e-05, 1e6), nu=1.5)
kernel = ConstantKernel(1.0) * (kernel_1 + kernel_2)

# GP definition and training
gaussian_process = GaussianProcessRegressor(
        kernel=kernel,
        optimizer="fmin_l_bfgs_b",
        n_restarts_optimizer=40
        )
gaussian_process.fit(train_ins_final, train_dr)

# Prediction ...
```

5.2.1 Area Profile Prediction

Initially, an attempt was made to predict the entire area profile of the deformed curve. To simplify the computational load, it was decided arbitrarily to predict it every 100 [mm], although this quantity can be modified in the *05_POST_PROC.ipynb* file. This operation aims to reduce the computational burden for the training time, by sacrificing precision

The implementation attempts followed three approaches:

- GPR *Cut* logic approach
- GPR *Black Box* logic approach
- Multi-layer perceptron MLP

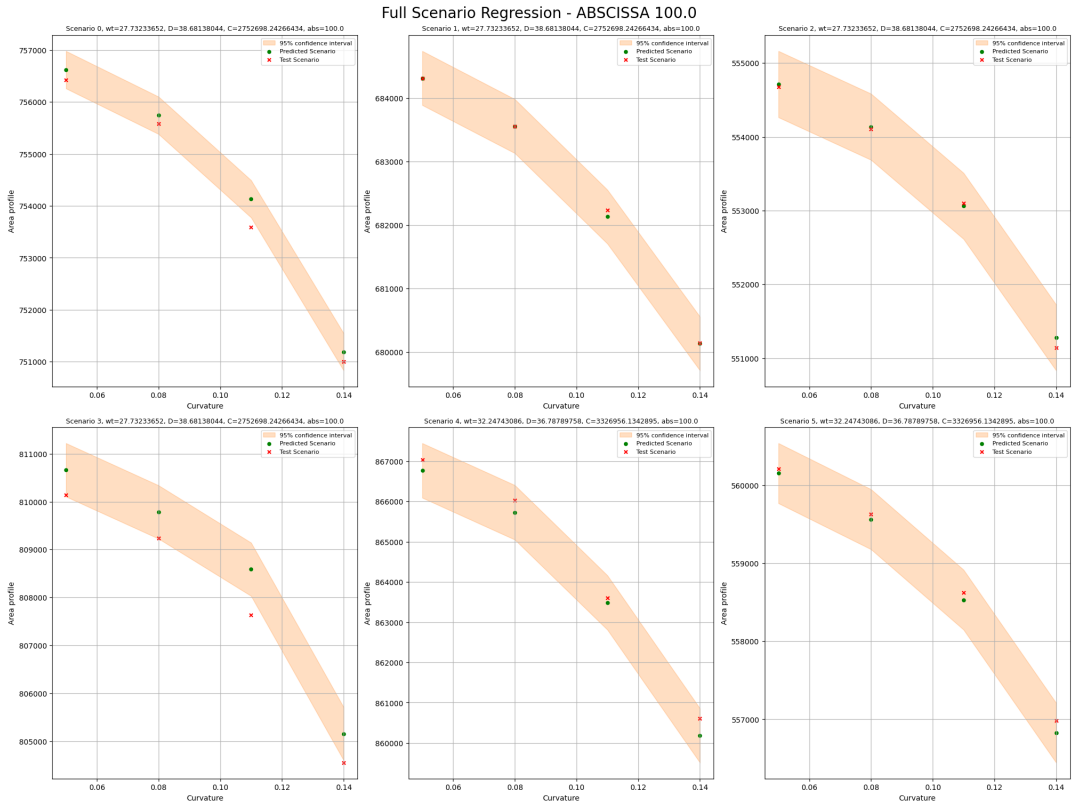


Figure 5.4: Example of a GPR prediction for the area value at a single axial coordinate, plotted against curvature for multiple scenarios. This illustrates the "cut" framework logic.

Cut

The first *Cut* approach aims to "slice" each curve at each curvilinear abscissa and predict the new value section by section. Figure 5.4, 5.5 show the results on this approach.
The predictions are quite accurate, as shown in Table 5.1, but the confidence interval tends to widen at higher abscissa, Figure 5.5.

Black box

The second method aims to build a prediction using each available curve as a single data point, without grouping by scenario. Each time, one curve is removed from the set, and an attempt is made to predict it, following the Leave-One-Out approach (Figure 5.6, Table 5.2).

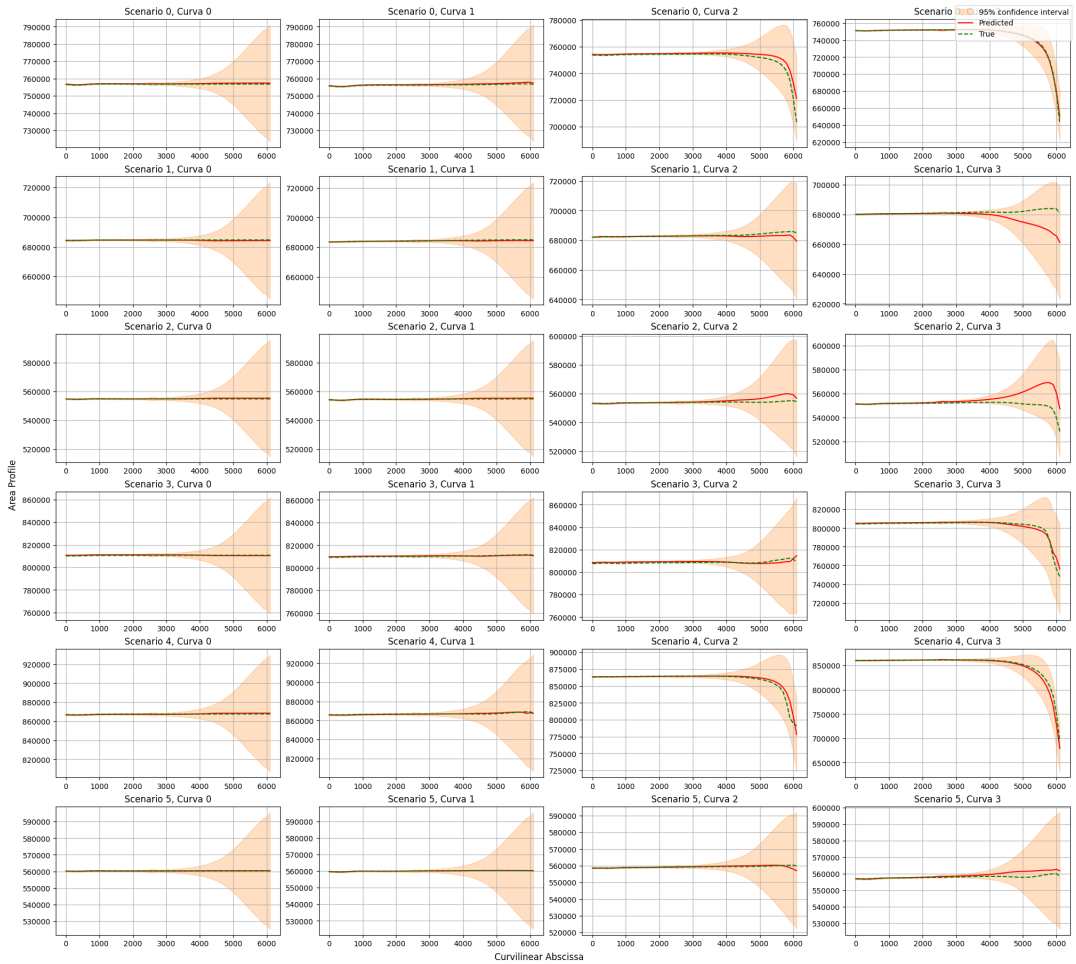


Figure 5.5: Area Profile Prediction with "Cut" framework with confidence interval

Table 5.1: Error metrics for the Area Profile (AP) prediction with the GPR "Cut" model, evaluated using Leave-One-Out Cross-Validation.

Error Metric	**Value**
Mean Squared Error (MSE)	7282354.0467
Root Mean Squared Error (RMSE)	2698.5837
Mean Absolute Error (MAE)	949.8000
R^2 Coefficient	0.9995

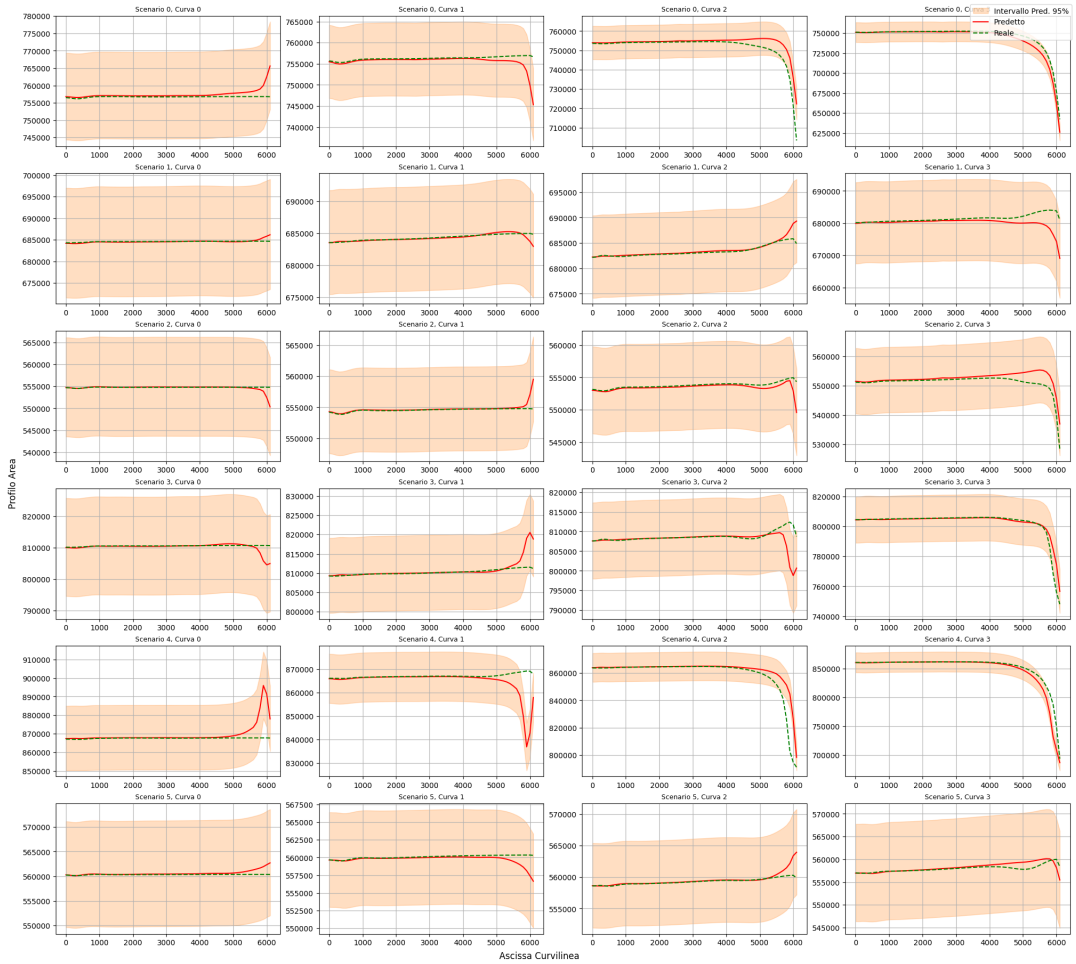


Figure 5.6: Area Profile Prediction with "Black Box" framework with confidence interval

Predicting the entire area profile, a functional regression task, is computationally demanding, both in dataset processing and model training.

Furthermore, achieving high accuracy across the entire vector output often requires a very large training dataset. A more pragmatic and widely adopted strategy in surrogate modeling is to focus on predicting specific scalar *Quantities of Interest (QoI)* that effectively summarize the system's behavior. Consequently, this study shifts focus to predicting two key performance metrics: Area Reduction (AR) and Diameter Reduction (DR).
These scalar values represent the maximum deformation of the Hollow Element and serve as critical indicators of structural collapse. Their practical relevance is

Table 5.2: Error metrics for the Area Profile (AP) prediction with the GPR "Black Box" model, evaluated using Leave-One-Out Cross-Validation.

Error Metric	**Value**
Mean Squared Error (MSE)	14027018.8618
Root Mean Squared Error (RMSE)	3745.2662
Mean Absolute Error (MAE)	1087.9382
R^2 Coefficient	0.9990

significant; for instance, in transport, AR and DR directly influence the change in flow capacity. In packaging applications, these metrics can determine whether a deformed container meets manufacturing tolerances and remains viable for sale. By focusing on AR and DR, we develop an efficient surrogate model targeted at the most critical metrics for the component's functional and structural evaluation.

5.2.2 DR - AR Prediction

The model's validation was performed using a Leave-One-Out Cross-Validation (LOOCV) procedure. The results of the initial GPR model, shown in Figure 5.7, while capturing the general trend, had 95% confidence intervals that extended to negative values, which are physically impossible for a diameter reduction.

To address this issue, a common technique for handling strictly positive data was employed: the model was retrained to predict a transformation of the output variable, specifically $\log(DR + 1)$. This transformation maps the positive target space to the real number line, aligning it with the Gaussian assumptions of the GPR model. As shown in Figure 5.8, this refined approach not only yields more accurate predictions but also produces narrower, physically consistent confidence intervals. The logarithmic transformation effectively linearizes the exponential nature of the collapse, significantly improving the surrogate's overall performance and reliability. The error metrics for the results on the test set, illustrated in Figure 5.8, are reported in table 5.3.

Prediction on new data

Interpolation The true effectiveness of the metamodel lies in its ability to predict the behavior of non simulated configurations. Figures 5.9b and 5.10b show

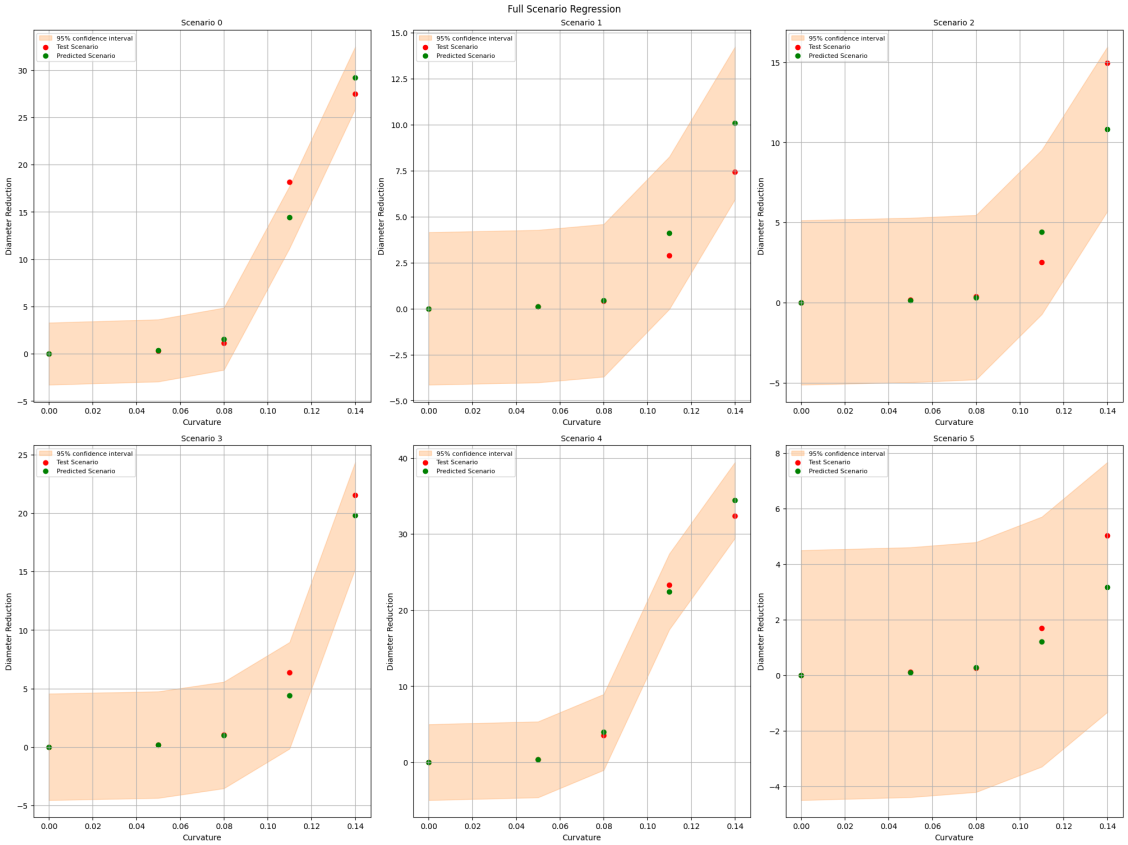


Figure 5.7: Predictions of Diameter Reduction (DR) using the basic GPR model, evaluated with a LOOCV procedure. The 95% confidence intervals extend to negative values, a physically implausible artifact for this metric.

Table 5.3: Error metrics for the Diameter Reduction (DR) prediction, obtained from the GPR model with logarithmic transformation and validated via LOOCV.

Error Metric	**Value**
Mean Squared Error (MSE)	4.3462
Root Mean Squared Error (RMSE)	2.0848
Mean Absolute Error (MAE)	1.0642
R^2 Coefficient	0.9486

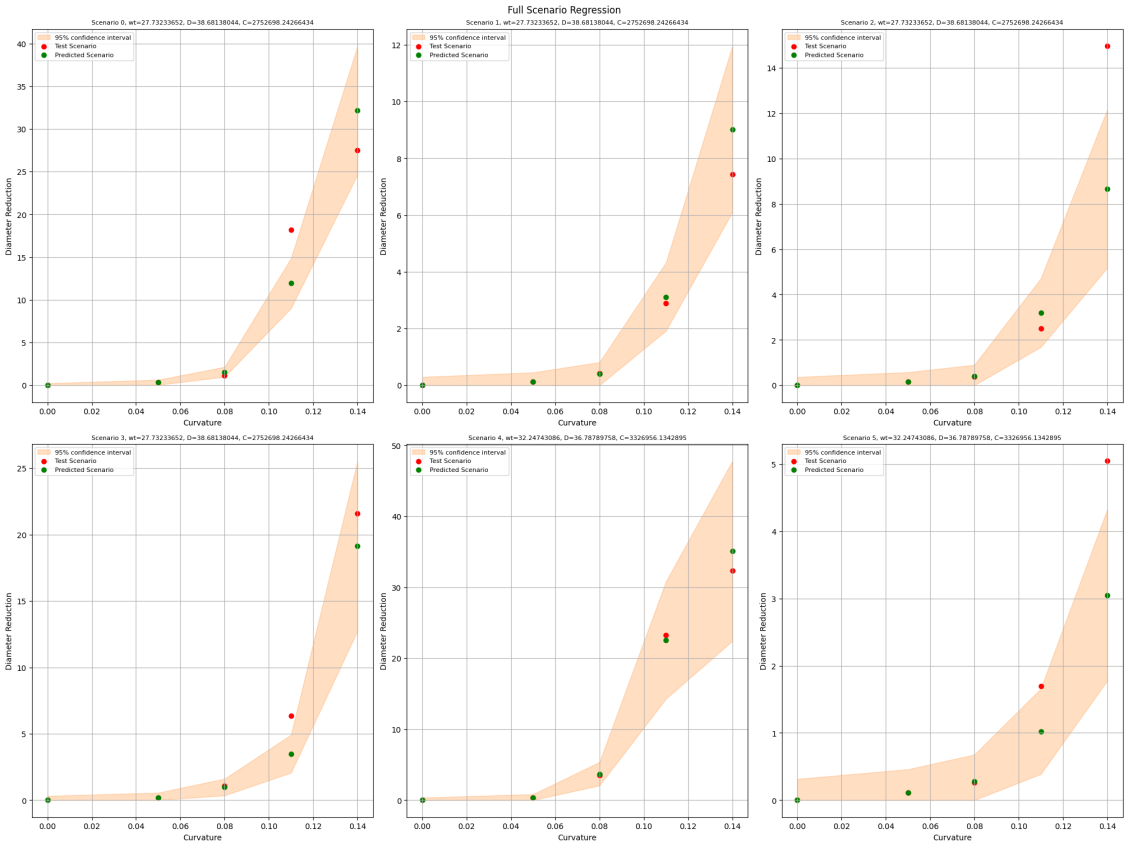


Figure 5.8: Predictions of Diameter Reduction (DR) using the GPR model with a logarithmic transformation of the output variable, log(DR+1). The transformation corrects the artifact of the previous model, yielding physically consistent confidence intervals.

predictions for two scenarios whose parameters lie within the training domain. The model provides physically plausible DR vs. Curvature response curves with relatively narrow confidence intervals. This indicates a high reliability of the model in the interpolation regime, thanks to a sufficient density of data in these regions of the parameter space.

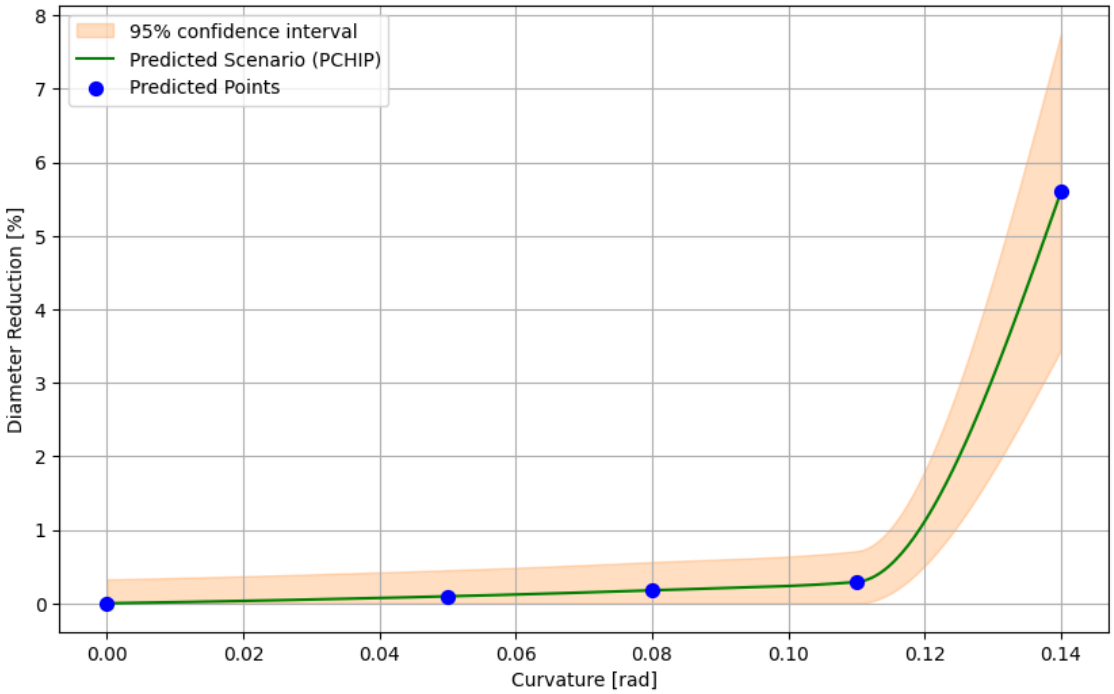
Extrapolation A configuration with an outer diameter (`OUTER_DIAMETER = 50`) outside the training range [32-42] was tested. As expected from the theory of Gaussian Processes, and shown in Figure 5.11b, the model produces a prediction but correctly signals low reliability through an extremely wide confidence interval. This should not be considered a weakness, but an intrinsic and fundamental feature of GPR, namely to quantify its predictive uncertainty in underexplored regions of the space.

Observations Being computationally almost instantaneous, the GPR metamodel is an ideal tool for the interactive exploration of the design space. The analysis of different configurations confirms that *Curvature* and *Wall Thickness* are the dominant parameters. The surrogate model allows for the rapid visualization and quantification of their impact, showing how a reduction in thickness has a progressively more marked effect as curvature increases, revealing a nonlinear interaction between the two parameters that the initial correlation analysis could not capture.

Should be stressed that the core of interest in the predictions lies in the range under 8/10 % for the DR: after that limit the structure is very compromised, and taken as collapsed.

(a) Input parameters for the interpolation test scenario 1.

Parameter	Value	Unit
Outer Diameter	35	mm
Wall Thickness	29	mm
Axial Load	1e6	N
External Pressure	1.5	MPa
Young Modulus	210000	MPa
Yield Stress 1	370	MPa
Yield Stress 2	470	MPa
EP2 Modulus	0.1	-
Ovality	0.001	-

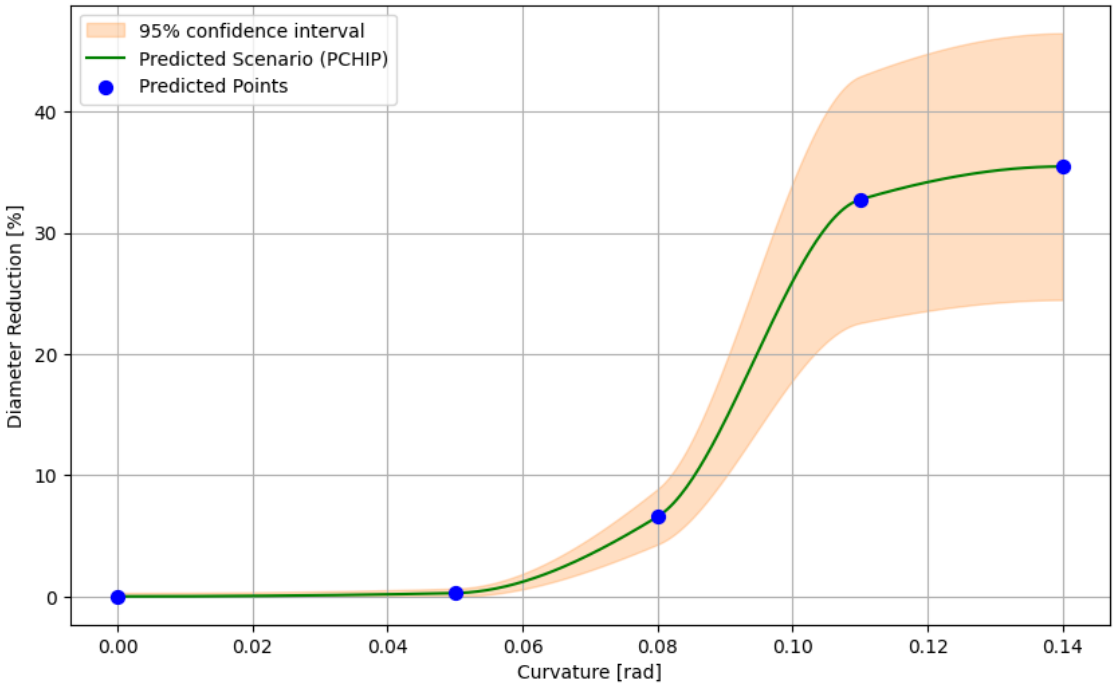


(b) Prediction of the DR vs. Curvature curve for the scenario defined above.

Figure 5.9: GPR prediction and input parameters for test scenario 1 (interpolation regime).

(a) Input parameters for the interpolation test scenario 2.

Parameter	Value	Unit
Outer Diameter	40	mm
Wall Thickness	26	mm
Axial Load	5e6	N
External Pressure	1.5	MPa
Young Modulus	210000	MPa
Yield Stress 1	380	MPa
Yield Stress 2	480	MPa
EP2 Modulus	0.15	-
Ovality	0.001	-

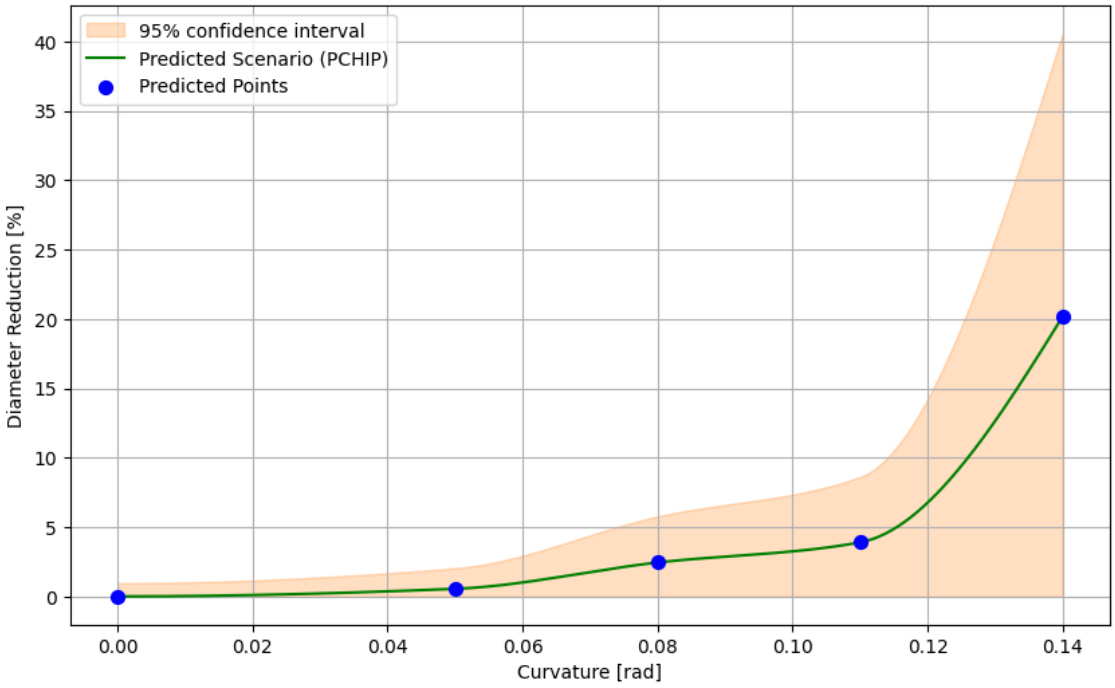


(b) Prediction for the second scenario in the interpolation regime.

Figure 5.10: GPR prediction and input parameters for test scenario 2 (interpolation regime).

(a) Input parameters for the extrapolation test scenario 3.

Parameter	Value	Unit
Outer Diameter	50	mm
Wall Thickness	29	mm
Axial Load	1e6	N
External Pressure	1.5	MPa
Young Modulus	210000	MPa
Yield Stress 1	370	MPa
Yield Stress 2	470	MPa
EP2 Modulus	0.1	-
Ovality	0.001	-



(b) Prediction for the scenario whose parameters are outside the training domain.

Figure 5.11: GPR prediction and input parameters for test scenario 3 (extrapolation regime). The model correctly signals high uncertainty when predicting outside the training domain.

5.3 MLP Metamodel

The training of the MLP models was conducted by minimizing the Mean Squared Error (MSE) loss function, a standard choice for regression problems. The Adam optimizer was employed with a learning rate of $\alpha = 0.001$, selected for its adaptive learning rate capabilities and robust convergence properties. The training was performed for 200 epochs. To mitigate overfitting and ensure generalization, a validation set was used to monitor performance; the model exhibiting the lowest validation error during the training process was saved and used for all subsequent predictions. This procedure was applied consistently for both the area profile and the scalar DR/AR prediction models.

5.3.1 Area Profile prediction

The third approach for this vectorial prediction aims for a more generic method: approximating the trend of the area profile using a fully connected Multi-Layer Perceptron Neural Network, trained to make a prediction on the new curve.

The input layer receives 11 features, comprising the 10 scenario parameters and the specific curvature value.

The `ProfileMLP` model is a Multi-Layer Perceptron, whose architecture is defined by a sequence of layers, where the output of one layer becomes the input of the next.

Table 5.4 shows the Architecture of the Multi-Layer Perceptron (MLP) used for the area profile prediction. For each layer, the type, activation function, and input/output feature dimensions are reported.

Table 5.4: Architecture of MLP for Area Profile predictions

Layer	Type	Activation	in_features	out_features
1	Linear	ReLU	11	128
2	Linear	ReLU	128	128
3	Linear	ReLU	128	64
4	Linear	ReLU	64	32
5	Linear	None	32	1

Listing 5.2: Definition of the MLP class in PyTorch.

```
import torch.nn as nn

class MLP(nn.Module):
def __init__(self, input_size, output_size):
super(MLP, self).__init__()
self.layer1 = nn.Linear(input_size, 64)
self.layer2 = nn.Linear(64, 32)
self.output_layer = nn.Linear(32, output_size)
self.relu = nn.ReLU()

def forward(self, x):
x = self.relu(self.layer1(x))
x = self.relu(self.layer2(x))
x = self.output_layer(x)
return x
```

Description

- **Input Layer (0):** Receives 11 features and projects them into a 128-dimensional space.
- **Hidden Layers (2, 4, 6):** Process the features and reduce the dimensionality from 128 to 64 and then to 32.
- **Activation Functions (ReLU):** Introduce the necessary non-linearity to learn complex relationships.
- **Output Layer (8):** Reduces the final vector to a single value, making the model suitable for regression tasks. There is no activation function, which would be more appropriate for classification problems.

The training was performed for 200 epochs, selecting the model with the lowest validation error on the test curves to mitigate potential overfitting. Figure 5.12 shows the results on test set.

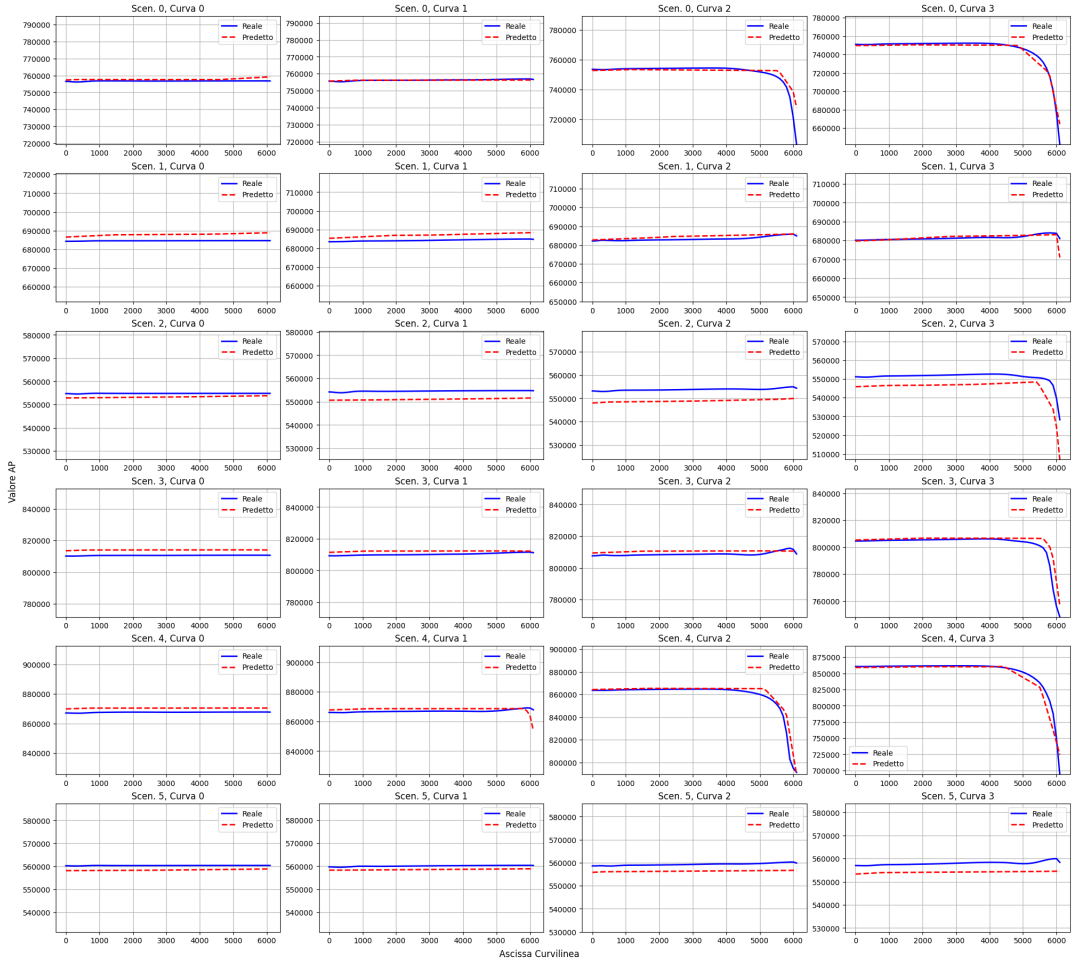


Figure 5.12: Comparison of true (solid blue line) and predicted (dashed red line) area profiles using the MLP framework for multiple test scenarios.

Table 5.5: Error metrics of the MLP model for the area profile prediction, calculated on the test set.

Metric	**Value**
Mean Squared Error (MSE)	12367506.0000
Root Mean Squared Error (RMSE)	3516.7465
Mean Absolute Error (MAE)	2501.2275
R^2 Coefficient	0.9991

The model's performance are comparable with the GPR, demonstrated by a coefficient of determination (R^2) of 0.9991. The absolute error metrics (Table 5.5) confirm this high precision.

5.3.2 DR prediction

The architecture for predicting the scalar DR value is a simpler MLP, detailed in Table 5.6. This is due to the simpler behavior of the DR curve with respect to the AP one.

Table 5.6: Architecture of MLP for Diameter Reduction predictions

Layer	Type	Activation	in_features	out_features
1	Linear	ReLU	10	64
2	Linear	ReLU	64	32
3	Linear	None	32	1

The architecture features two hidden layers with an initial expansion to 64 nodes and a subsequent reduction to 32 nodes, before converging to a single output neuron. Note that the `ReLU` activation function is applied between the layers. Figure 5.13 shows the results on test set.

The model's performance was evaluated on a test set, with the metrics reported in Table 5.7. The values indicate a good predictive capability.

Table 5.7: Error metrics of the MLP model for the Diameter Reduction (DR) prediction, calculated on the test set.

Metric	Value
Mean Squared Error (MSE)	3.1209
Root Mean Squared Error (RMSE)	1.7666
Mean Absolute Error (MAE)	1.2113
R^2 Coefficient	0.9673

5.3.3 AR prediction

The same MLP architecture was used for the Area Reduction prediction. The results are shown in Figure 5.14 and Table 5.8.

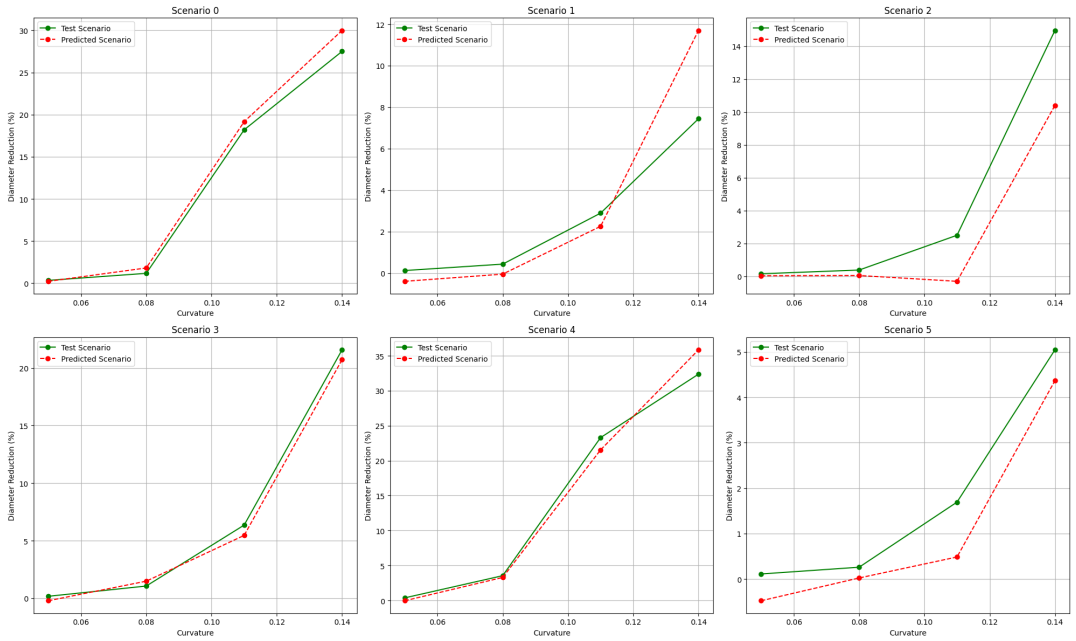


Figure 5.13: Comparison between the true Diameter Reduction (DR) values (solid green line) and the MLP model's predictions (dashed red line) for six test scenarios, as a function of the imposed curvature.

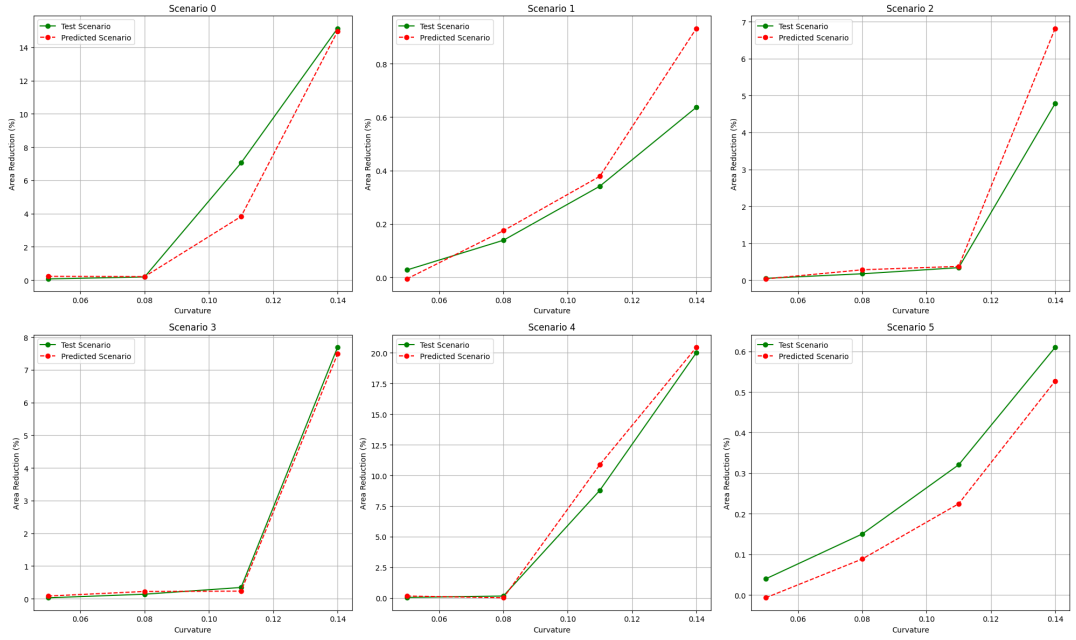


Figure 5.14: Comparison between the true Area Reduction (AR) values (solid green line) and the MLP model's predictions (dashed red line) for six test scenarios, as a function of the imposed curvature.

Table 5.8: Error metrics of the MLP model for the Area Reduction (AR) prediction, calculated on the test set.

Metric	Value
Mean Squared Error (MSE)	0.8073
Root Mean Squared Error (RMSE)	0.8985
Mean Absolute Error (MAE)	0.4039
R^2 Coefficient	0.9703

A notable drawback of the MLP approach is its sensitivity to initialization, with performance metrics varying between training runs; the R^2 coefficient, for instance, could occasionally drop as low as 0.90. This stochastic nature, combined with the model's inherent inability to provide predictive confidence intervals, leads to the conclusion that the GPR model is superior for this engineering application, where both robustness and uncertainty quantification are paramount.

To enhance stability and accuracy, a hybrid model could an optimal approach. The final solution is a weighted ensemble of the outputs from three constituent models. The weights are calibrated based on each model's performance metrics or the inverse of its confidence interval's width. This methodology effectively leverages the strengths and mitigates the limitations of each individual approach.

5.4 Advantages and Limitations of the Surrogate Model

Advantages The primary advantage of the surrogate model lies in its extraordinary computational efficiency. A prediction that requires a fraction of a second replaces a full FEM simulation that would otherwise take hours, thereby enabling a nearly exhaustive exploration of the parameter space. This speed transforms the design process, allowing for the rapid identification of critical regions and the predictions. Beyond pure speed, the GPR based model offers a crucial benefit for engineering applications: the intrinsic quantification of uncertainty. By providing a confidence interval with each prediction, the model does not just estimate a result but also measures its own reliability, a fundamental element for supporting robust

decision-making processes.

Limitations However, this methodology is subject to inherent limitations and trade offs. The main benefit of efficiency is counterbalanced by a significant initial computational investment required to generate the FEM training dataset, which for this study amounted to approximately 10-12 CPU-hours for a recent 4-core PC with 16GB of RAM. The final model's accuracy is intrinsically linked to the quality and density of this initial sampling; a poorly designed Design of Experiments (DoE) would irreparably compromise the surrogate's performance. Furthermore, the model operates as a sophisticated interpolator and does not "understand" the underlying physics. Consequently, as has been demonstrated, it loses accuracy and reliability outside the domain of the training data (extrapolation), having learned a functional relationship that is only valid locally. Finally, scalability challenges arise for more complex problems: while the performance of standard GPR models can degrade in much higher dimensional spaces due to the "curse of dimensionality," alternative approaches like Neural Networks tend to suffer in data limited settings, where they are prone to overfitting.

Improve performance The predictive performance of the surrogate framework can be further enhanced through several strategic refinements. Primarily, expanding the training dataset by executing additional high-fidelity FEM simulations would provide a denser sampling of the parameter space, enabling the model to capture more complex nonlinearities and reduce interpolation uncertainty.
For the specific task of Area Profile (AP) prediction, increasing the sampling resolution along the element's axial coordinate would yield a more detailed vector representation of the collapse, potentially improving the accuracy of the functional regression.
Moreover, the implementation of hybrid or ensemble models, as previously suggested, could combine the probabilistic strengths of GPR with the nonlinear mapping capabilities of MLP to produce more robust predictions.
Finally, a more rigorous feature selection process could simplify the problem; for instance, if a parameter like the Young's Modulus is constant for a given material class, fixing its value would reduce the input space dimensionality, allowing the model to focus on more influential variables.

It is crucial, however, to acknowledge that while these strategies can tune the model's accuracy to meet specific engineering requirements, the surrogate remains an approximation. Its predictive fidelity will always be fundamentally bounded by, and consequently inferior to, that of the underlying Finite Element model it emulates.

6

Industrial Impact and Applications

6.1 Motivations for Industrial Adoption

The primary motivation for industrial adoption of the framework developed in this thesis is the reduction of the computational cost of high-fidelity nonlinear simulations. Finite Element Analysis (FEA) remains the reference method for predicting structural collapse, but its runtime limits its direct application in parametric studies and design optimization.

The surrogate models developed in this work, based on Gaussian Process Regression and Neural Networks, provide predictions in negligible computational time, once trained. This enables their integration into design workflows and allows real-time exploration of design alternatives.

It must be noted, however, that the adoption of surrogate models also presents practical limitations. The availability of high-quality FEM datasets, the limited reliability of extrapolation outside the training domain, and the trade-off between training effort and prediction accuracy remain critical aspects for robust deployment. This extrapolation limitation can be mitigated by Physics-Informed Neural Networks (PINNs). By embedding the governing physical equations directly into the model's loss function, PINNs are forced to generate solutions that are physi-

cally consistent, even in data-sparse regions. This provides a more robust basis for extrapolation compared to purely data-driven approaches.

6.2 Quantified Impact on Hollow-body Collapse Prediction

The main advantage of the framework is the measurable increase in computational efficiency.
A single high-fidelity FEM simulation required between 80 and 700 seconds, while the trained surrogate model delivers predictions in about 0.1 seconds per configuration. This corresponds to a computational speed-up of 3 to 4 orders of magnitude, enabling parametric sweeps of 10^3 to 10^4 configurations.
For instance, this enables a workflow where the computationally expensive training set is generated in advance, at the beginning of a product's design phase. The resulting surrogate model is then available for near-instantaneous analysis during subsequent design iterations, yielding significant time savings.

This computational gain enables broader parametric studies and faster convergence to optimal designs:

- *Reduction of Computational Costs*: The framework drastically lowers the computational budget required for an extensive design campaign.
- *Acceleration of the Design Cycle*: The surrogate model facilitates the rapid exploration of vast parameter spaces, enabling thousands of design variations to be evaluated in minutes rather than weeks. This supports the identification of non-intuitive, highly optimized solutions and significantly shortens the product's time-to-market by compressing the analysis and iteration phases.
- *Improved Risk Assessment*: The model's efficiency makes comprehensive safety and reliability analyses computationally tractable. Furthermore, these evaluations can be performed on low-cost, low-power hardware, democratizing access to advanced structural analysis, available in a wide range of applications.
- *Product Cost Reduction*: The advanced optimization capabilities unlocked by the surrogate model lead to cost savings in the physical product. By exhaustively exploring the design space, the framework can identify configurations

that meet performance targets while minimizing material volume or simplifying manufacturing processes, thereby directly reducing production costs.

- *Hybrid Multi-fidelity and Data Integration*: The framework is architected to support the fusion of high-fidelity simulation data with empirical measurements from physical tests. This multi-fidelity approach produces a surrogate model that is not only theoretically sound but also calibrated against real-world performance, where available. The result is a more robust and reliable predictive tool that corrects for inherent discrepancies between the numerical model and physical reality.

6.3 Comparative Examples from the Literature

To place this work within the state-of-the-art, this section compares the developed framework with two representative studies. The objective is to highlight both the methodological analogies and the distinctive contributions of this thesis.

6.3.1 Coronary Stent Optimization via Kriging

Li et al. [29] applied Kriging (Gaussian Process) to optimize coronary stent geometry, with the aim of reducing undesirable phenomena such as *dogboning* during expansion. Their workflow involved building a limited set of FE models, defining scalar Quantities of Interest (QoIs), and using an Expected Improvement criterion in an adaptive loop (Efficient Global Optimization).

While the general paradigm (FEA $\rightarrow$ Surrogate Model $\rightarrow$ Optimization) is shared, the present framework differs in scope. Stent optimization focused on scalar QoIs, while this work also predicts vector-valued profiles (the Area Profile, AP), providing a richer description of the structural response. In addition, the dataset here was generated via a Sobol one-shot Design of Experiments, rather than adaptively, enabling systematic exploration of nonlinear interactions leading to buckling, at the cost of a larger initial dataset. This methodological distinction stems from the divergent objectives of the two studies. The former targeted a specific optimization

Table 6.1: Comparison between reference works and this thesis.

Metric	Li et al. (2013)	Shaikh et al. (2024)	This Work
Application Domain	Medical Stents	EV Battery Crash	Structural Collapse
Surrogate Model	Kriging	GPR	GPR & MLP
Avg. FE runtime	Not stated	High (crash sim.)	2–12 min
Surrogate runtime	Near-instantaneous	Near-instantaneous	~ 0.1 s
Output Type	Scalar QoIs	Scalar QoIs	QoIs & Vectorial Profiles
Key Contribution	Iterative optimization	UQ on large datasets	Automated Pipeline

problem, for which an adaptive sampling strategy is efficient, while the present work is focused on a comprehensive parameter space exploration, where a space-filling design is required to capture global behavior and parameter interactions.

6.3.2 Probabilistic Surrogates for Crashworthiness

Shaikh et al. [30] studied the crash performance of battery enclosures for electric vehicles using a Gaussian Process Regression surrogate trained on a high-throughput FE campaign. Their work emphasized uncertainty quantification, using predictive variance for Monte Carlo propagation of uncertainties. This illustrates the role of probabilistic surrogates in safety-critical design.
In contrast, the present work applies uncertainty quantification (UQ) to assess prediction reliability in a multidimensional parameter space, with fewer FE runs but broader scope, including scalar and vectorial outputs. Automation and reproducibility of the pipeline represent distinctive aspects compared to Shaikh's workflow.

Convergences and Differences

This comparison shows that surrogate modeling is already consolidated in related fields. The contribution of this thesis is to extend the methodology from predicting scalar indicators to approximating full structural response profiles with quantified uncertainty, embedded in a full automated pipeline.

6.4 Transferability Example: Planar Antenna Optimization

To illustrate the versatility of the framework, this section outlines a possible extension to a different domain: optimization of a planar patch antenna for mobile devices. This is proposed as a proof of transferability rather than a developed contribution. The problem differs in physics (electromagnetics instead of mechanics) but retains the challenge of costly high-fidelity simulations.

6.4.1 Problem Definition and Methodology

The antenna geometry can be parameterized by variables such as *Patch Length (L_p)*, *Patch Width (W_p)*, and *Substrate Height (h)*. Performance is typically evaluated through computationally intensive full-wave electromagnetic simulations (e.g., CST Studio Suite), making exhaustive exploration often infeasible, even with this simple geometry (Figure 6.1).

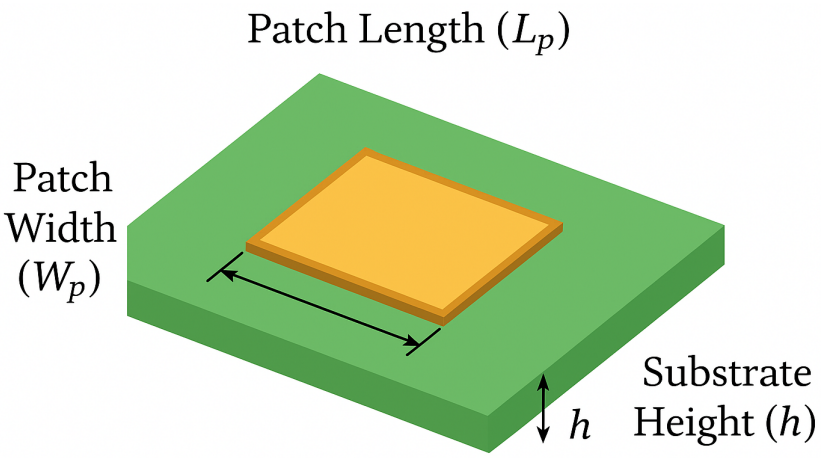


Figure 6.1: Example of patch antenna geometry to be optimized.

The optimization would follow a multi-objective strategy with two requirements:

- Peak Gain (G), to be maximized.
- Operational Bandwidth (BW), to be maximized.

The methodology would adapt the framework as follows:

1. **Dataset Generation:** Sobol sampling is used to generate antenna geometries. Each configuration is simulated to obtain G and BW.

2. **Metamodel Training:** Two GPR surrogates are trained, one for G and one for BW, with predictive variance used to assess reliability.

3. **Optimization:** The surrogates are used as inexpensive objective functions in a multi-objective optimization algorithm.

6.4.2 Expected Outcomes and Value Proposition

The expected outcome is a predictive tool enabling real-time exploration of antenna designs. Instead of long simulations, engineers could perform near-instantaneous evaluations, including sensitivity analyses such as that conceptually illustrated in Figure 6.2, where the effect of patch length on gain is estimated together with confidence intervals.

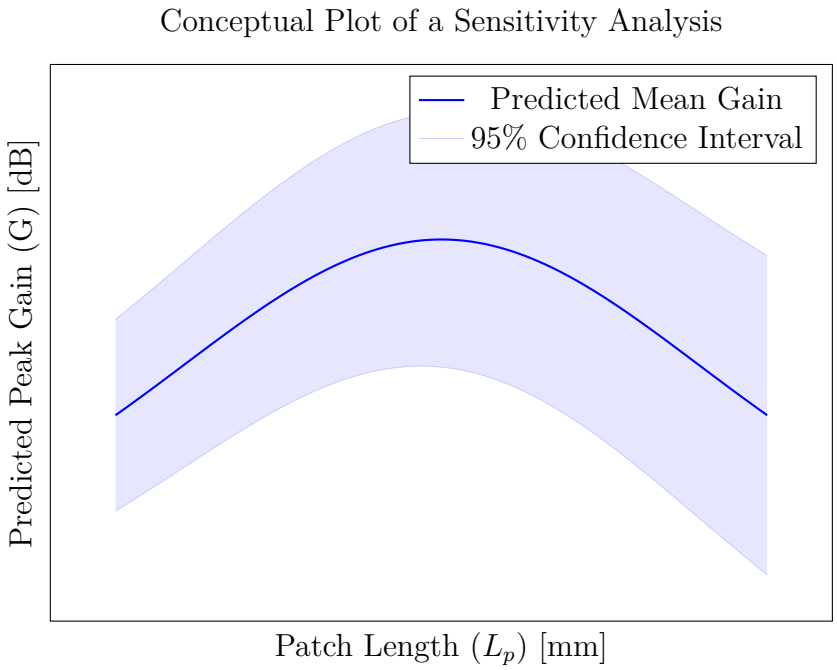


Figure 6.2: Qualitative example of representation GPR-based sensitivity analysis: predicted mean and 95% confidence interval. (Data are purely arbitrarily generated)

The optimal design configuration for a component is rarely universal: it is highly dependent on application-specific constraints such as loading conditions, material costs, and manufacturing limitations. A key advantage of the surrogate model framework is its reusability. Once trained on the general parameter space, the same model can be used to rapidly determine optimal solutions for a wide range of different constraint sets, without the need for retraining. This ”train-once, optimize-many” paradigm is particularly valuable for specialized companies that focus on a narrow portfolio of products but must deliver highly customized, high-performance engineering solutions for each client or use case.

A final compromise design would be validated with a single high-fidelity simulation, expected to fall within the confidence bounds predicted by the surrogate.

7 Conclusions

This thesis presented the development of an automated pipeline for the surrogate-based prediction of collapse behavior in hollow structural elements. The workflow integrates Design of Experiments (Sobol sampling), high-fidelity finite element simulations (Abaqus), systematic extraction of collapse indicators, and the construction of surrogate models (Gaussian Process Regression and Multi-Layer Perceptrons). A key contribution lies in extending the prediction beyond scalar quantities of interest (such as AR and DR) to full vector-valued responses, namely the Area Profile (AP), thus enabling a more detailed description of collapse mechanisms.

The results show that the proposed surrogates achieve substantial computational gains: from simulation times of 80–700 seconds per case to surrogate predictions of the order of 0.1 seconds, i.e., a speed-up of three to four orders of magnitude. While Gaussian Processes provide predictive uncertainty and remain competitive on smaller datasets, neural networks are better suited for handling larger and more complex data structures. Both approaches, however, demonstrate errors within an acceptable engineering range.

Several limitations emerged. The reliance on artificial stabilization in the FE solver, although practical, requires further sensitivity analyses to ensure that damping artifacts do not bias the collapse indicators. Likewise, the current validation strategy should be expanded beyond leave-one-out for 1D cases, incorporating explicit train/test splits and coverage metrics for predictive uncertainty. These aspects

are essential before industrial adoption.

Overall, the work demonstrates that surrogate models can effectively accelerate virtual design and optimization in crash-related applications. The pipeline is reproducible, extendable, and compatible with more advanced strategies, such as sequential sampling or multi-fidelity integration. Future developments should focus on improving calibration of predictive uncertainties, validating against experimental data, and embedding the surrogates directly into design workflows.

A

Buckling Analysis Procedure

This appendix outlines the standard computational workflow for the structural instability analysis of slim components, such as the pipe under study. The methodology is divided into two sequential and complementary phases: a linear eigenvalue buckling analysis to identify critical loads and mode shapes, followed by a nonlinear post-buckling analysis to simulate the real behavior of the structure, including its sensitivity to geometric imperfections. This approach is fundamental to distinguish between a collapse dominated by geometric instability (buckling) and one governed by material plasticization, whose numerical treatment is well-established in the literature on nonlinear finite element methods [1, 3].

A.1 Phase 1: Linear Eigenvalue Analysis

Linear buckling analysis is the first step in quantifying a structure's propensity for local instability. Global buckling is the bending of the entire member while local buckling is the wrinkling of a component of its cross-section. The focus is on the second, since is more critical in this problem.
The objective is to calculate the theoretical critical loads and the corresponding deformation shapes (instability modes) in a purely elastic regime, according to classical stability theory [6].

A.1.1 Objective

To determine the critical load multiplier λ and the associated instability mode shape ϕ.

A.1.2 Procedure

1. **Application of a Reference Load:** After a step where compression and external pressure are applied, a preliminary static analysis is performed by applying a nominal reference load, F_{ref}. In this study, this load corresponds to a imposed curvature of 0.01 radians. The analysis calculates the initial stress state required for the formulation of the geometric stiffness matrix $[K_{geo}(F_{ref})]$.

2. **Solving the Eigenvalue Problem:** The FEM solver (e.g., Abaqus [26]) is employed to solve the generalized eigenvalue problem, as discussed in Section 2.2.3:

$$([K_{mat}] + \lambda_i[K_{geo}])\{\phi_i\} = \{0\}$$

   where:

   - $[K_{mat}]$ is the conventional elastic stiffness matrix.
   - $[K_{geo}]$ is the geometric stiffness matrix, which depends on the pre-stress state induced by F_{ref}.
   - λ_i is the i-th eigenvalue, representing the critical load multiplier.
   - $\{\phi_i\}$ is the i-th eigenvector, which represents the shape of the instability mode (eigenmode).

3. **Calculation of the Critical Load:** The theoretical critical load for the i-th mode is given by:

$$F_{crit,i} = \lambda_i \cdot F_{ref}$$

   The first positive eigenvalue, λ_1, corresponds to the Euler critical load, which constitutes the theoretical upper bound for the onset of instability in a perfect structure.

4. **Identification of Relevant Modes:** The first N instability modes are extracted (e.g., N=20). The eigenvectors $\{\phi_i\}$ are visually inspected to identify those that represent physically realistic collapse mechanisms relevant to the subsequent nonlinear analysis. The first mode is identified as the main candidate.

A.2 Phase 2: Nonlinear Post-Buckling Analysis

Linear analysis is insufficient to describe the actual behavior of a structure, as it neglects material (plasticity) and geometric (large displacements) nonlinearities, as well as the crucial effect of geometric imperfections, which are ubiquitous in real components and can significantly reduce the load-bearing capacity.

A.2.1 Objective

To simulate the complete load-displacement path of the structure, capturing the post-buckling behavior and determining the actual collapse load and mechanism.

A.2.2 Procedure

1. **Introduction of a Geometric Imperfection:** To trigger a realistic post-buckling equilibrium path, a geometric imperfection is introduced into the model's mesh. The perfect geometry is perturbed by adding one or more mode shapes, $\{\phi_k\}$, obtained from the linear analysis, scaled by a small factor δ. The new nodal configuration $X_{imperfect}$ is:

   $$X_{imperfect} = X_{original} + \delta \cdot \{\phi_k\}$$

   The scale factor δ used is a small fraction of a characteristic dimension of the component, that is 10% of the wall thickness ($\delta = 0.1 \cdot t$).

2. **Incremental Nonlinear Analysis:** A nonlinear static analysis is performed, applying the load incrementally. A *Static General* analysis step was used, enhanced with an automatic stabilization technique based on energy dissipation. This method introduces a small amount of fictitious damping into the model, allowing the solver to overcome convergence difficulties at points of instability (bifurcation or limit points). This traces the post-buckling path while retaining the ability to perform the subsequent unloading analysis, which was critical for this study. Based on the user's values, the dissipated energy fraction was set to a low value (e.g., `0.01`) to minimize its impact on the physical results while ensuring numerical stability.

3. **Discrimination of the Collapse Mechanism:** By analyzing the results (e.g., load displacement curves and stress contours), it is possible to identify the

primary cause of collapse:

- **Collapse by Instability (Buckling):** The structure undergoes large deformations and loses its load resistance while stresses are still predominantly in the elastic range ($\sigma < \sigma_y$). The collapse is governed by geometric instability.
- **Collapse by Plasticization:** The structure reaches the material's yield stress ($\sigma \geq \sigma_y$) before significant instability phenomena occur. The collapse is governed by the material behavior and may manifest as localized plastic deformation, such as "denting".

The combination of these two analyses provides a complete and physically accurate framework for investigating structural collapse phenomena.

A.3 Illustrative Case Studies

To facilitate this analysis, two extreme configurations have been utilized. The objective of this approach is to ascertain whether buckling occurs in the model, or whether the structural collapse is solely attributable to material collapse.

A.3.1 Configuration 1: Instability-Dominated Collapse

The selection of parameters aims to create a geometrically unstable configuration by maximizing the diameter-thickness ratio and applying high compressive and pressure loads.

Preliminary Analysis

A preliminary nonlinear analysis using a *Static General* step failed to complete, as shown in Figure A.1. The structure exhibits a catastrophic, localized collapse with multiple folds, characteristic of buckling. The analysis log confirmed that numerous eigenvalues of the stiffness matrix became negative, indicating a loss of stability.

Table A.1: Parameters for Instability-Dominated Case

Parameter	Value	Unit
D_out	42	inches
Wall_Thickness	25	mm
Compression_Force	5	MN
External_pressure	2	MPa
SY1	375	MPa
SY2	450	MPa
EP2	0.15	-
Ovalisation	0.001	-
Curvature	0.1	$1/mm$

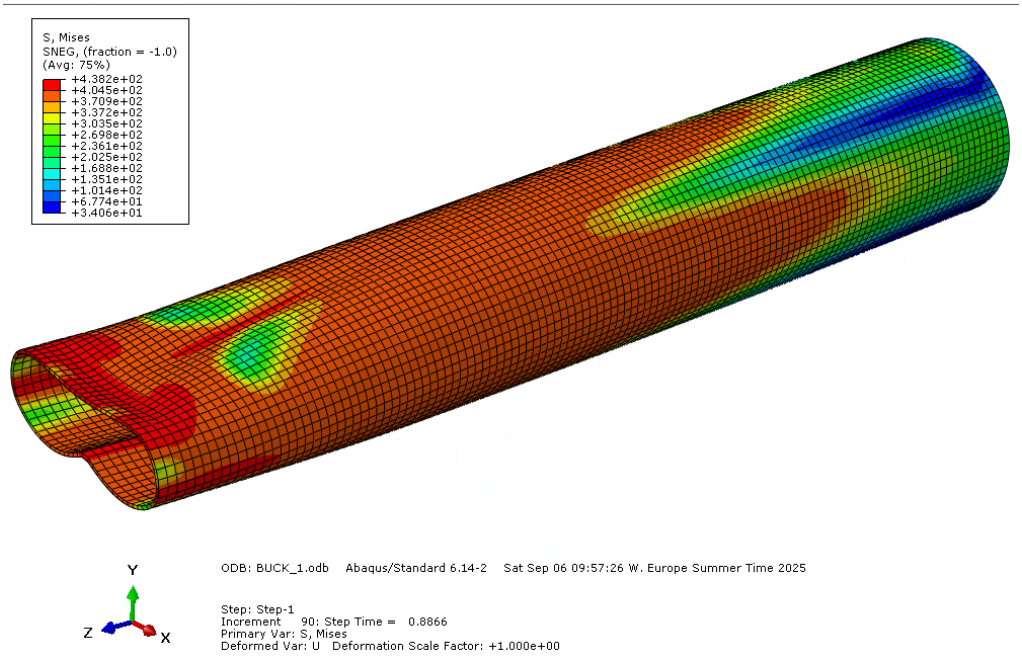


Figure A.1: Preliminary analysis of the instability-dominated case, showing collapse before completion.

Linear Eigenvalue Analysis

The linear analysis identified a first buckling mode with a low eigenvalue of **32.148**. So the first mode of instability occurs around 0.32 [rad] of imposed curvature, 2.3 times the maximum curvature scenario (0.14 [rad]).
The mode shape, shown in Figure A.2, is a characteristic spiral wrinkling pattern, confirming the structure's high susceptibility to buckling under the applied loads.

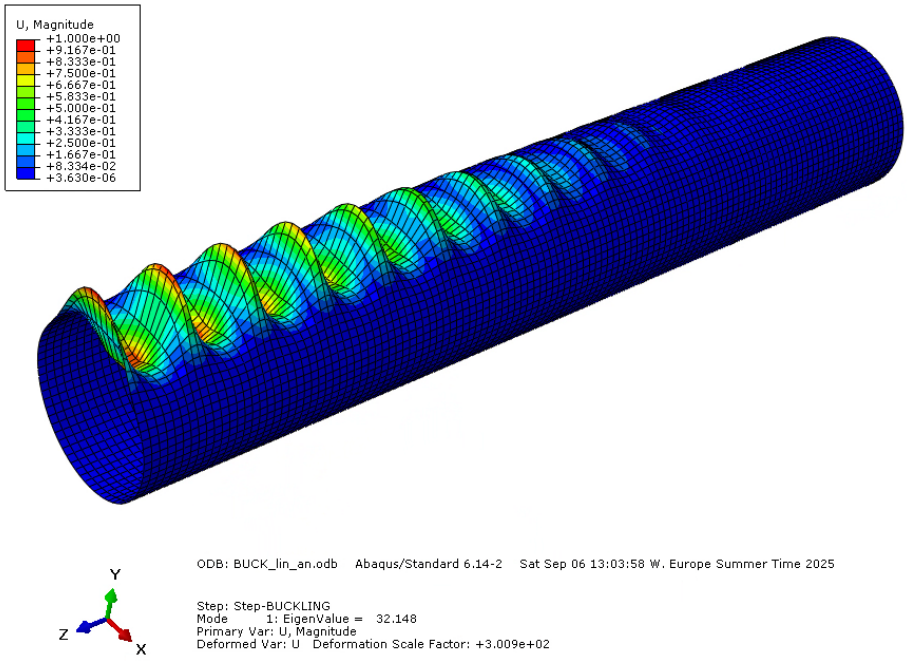


Figure A.2: First eigenmode from the linear buckling analysis for Configuration 1.

Nonlinear Post-Buckling Analysis

Introducing the first eigenmode as an initial imperfection, the nonlinear analysis successfully traced the post-buckling path. The final deformed shape (Figure A.3) closely resembles the preliminary analysis result, confirming that the collapse is triggered and governed by geometric instability.

A.3.2 Configuration 2: Material-Dominated Collapse

This configuration uses a smaller diameter-to-thickness ratio and lower loads to create a more stable geometry, where material failure is expected to precede global instability.

Preliminary Analysis

The preliminary analysis completed successfully, even with a significantly higher imposed curvature. Figure A.4 shows that the collapse is characterized by a large, smooth ”denting” deformation at one end. The stress plot indicates widespread

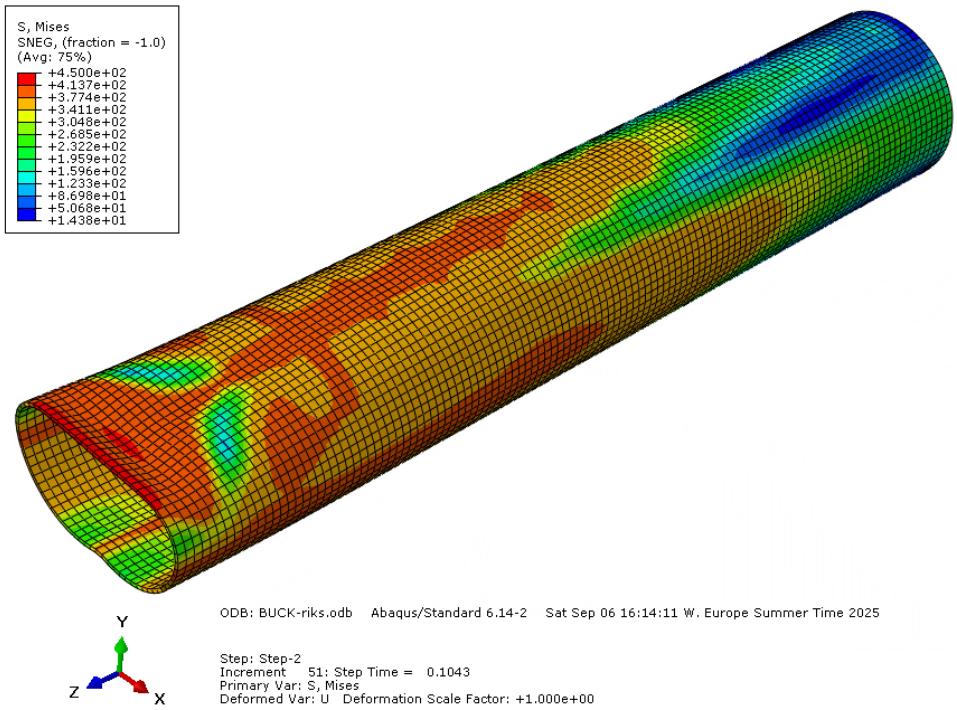


Figure A.3: Final deformed shape from the nonlinear post-buckling analysis for Configuration 1.

Table A.2: Parameters for Material-Dominated Case

Parameter	**Value**	**Unit**
D_out	32	inches
Wall_Thickness	35	mm
Compression_Force	1	MN
External_pressure	0.5	MPa
SY1	375	MPa
SY2	450	MPa
EP2	0.15	-
Ovalisation	0.001	-
Curvature	0.4	$1/mm$

plasticization across the section, suggesting that the material's yield strength was reached and exceeded, which is typical of a material-driven failure.

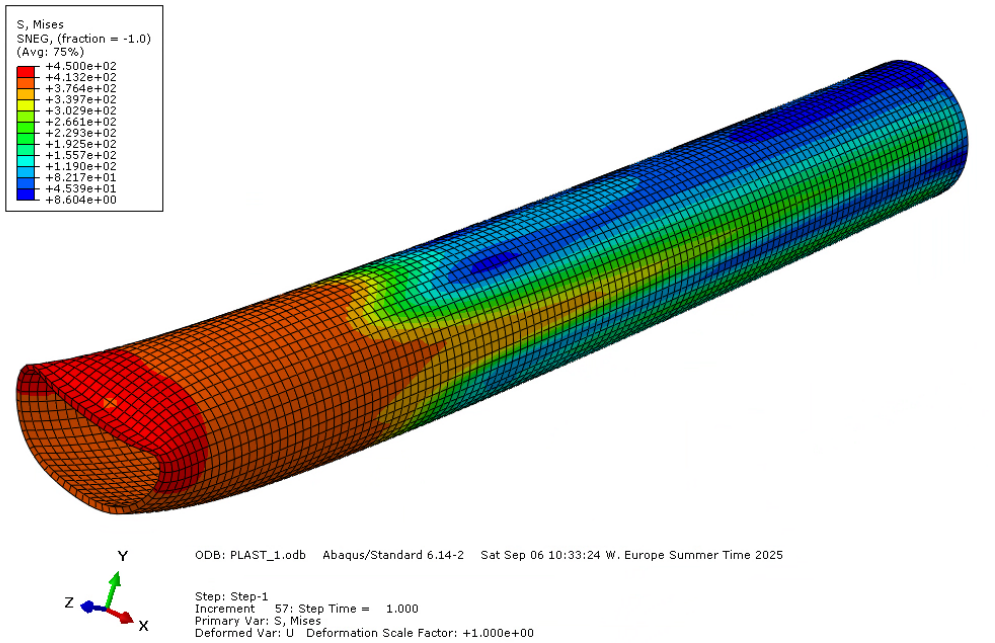


Figure A.4: Preliminary analysis of the material-dominated case, showing extensive plastic deformation.

Linear Eigenvalue Analysis

The linear buckling analysis for this configuration yielded a first mode eigenvalue of **79.035**. So the first mode of instability occurs at 0.79 [rad] of inposed curvature, 5.6 times the maximum curvature scenario (0.14 [rad]).
This value is more than double that of Configuration 1, indicating a much higher theoretical resistance to buckling. The mode shape is shown in Figure A.5.

Nonlinear Post-Buckling Analysis

The nonlinear analysis (Figure A.6) confirms the preliminary findings. The final collapse is a result of progressive material yielding leading to large, localized deformation. While geometric changes are significant, they are a consequence of plastic flow rather than an elastic instability trigger.

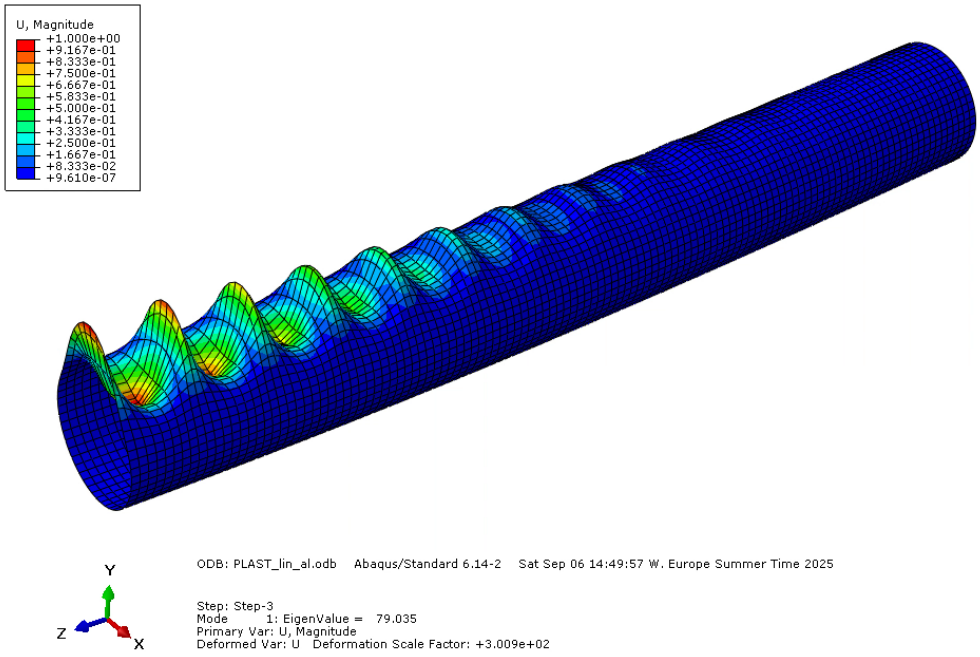


Figure A.5: First eigenmode from the linear buckling analysis for Configuration 2.

A.4 Conclusion on Collapse Mechanisms

The analysis of these two extreme cases demonstrates that structural collapse is rarely a pure phenomenon but rather a combination of geometric instability and material failure. However, the dominant mechanism can be clearly identified. In *Configuration 1*, the low eigenvalue and the resulting collapse morphology confirm that geometric instability (buckling) is the primary driver of failure. The structure loses instantaneously its ability to carry load due to a change in its geometry long before the material is fully exploited. Conversely, in *Configuration 2*, the high buckling eigenvalue and the widespread plastic deformation observed in the nonlinear analysis show that material collapse (plasticization) is the dominant failure mode. The structure collapses because the material yields under the applied loads, leading to large, irrecoverable deformations. The role of elastic instability is secondary but still observable. This dual analysis provides a robust framework for understanding and predicting the complex failure behaviors of such components.

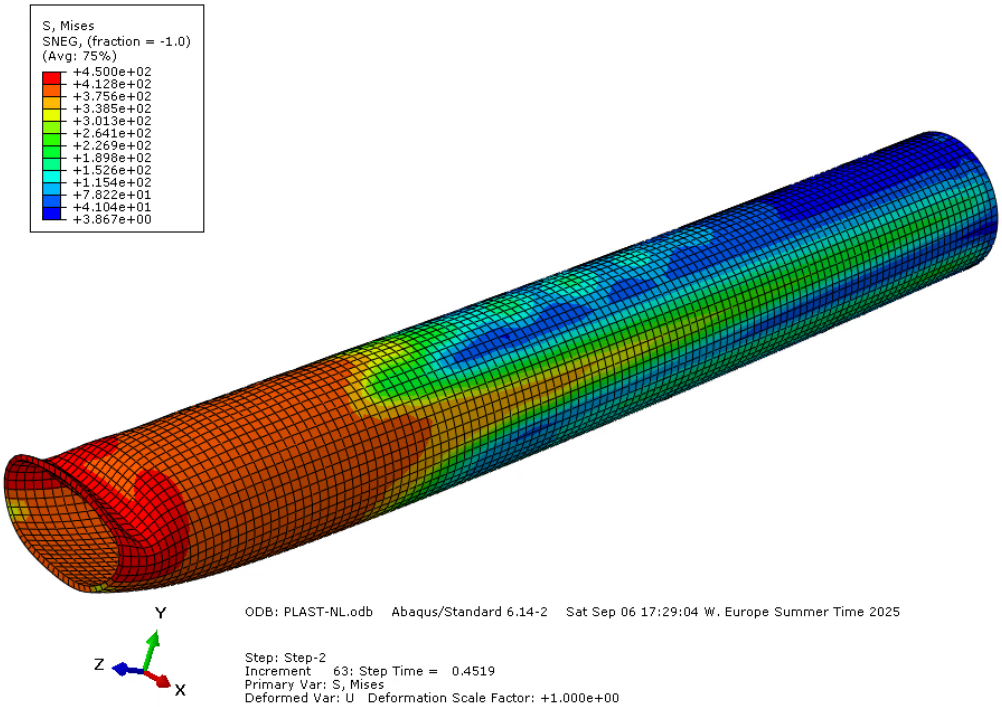


Figure A.6: Final deformed shape from the nonlinear post-buckling analysis for Configuration 2.

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The Gemini V2.5 Pro language model was utilized for stylistic refinement and code debugging.